

Corporate Sustainability Positioning and Conduct on TikTok

***How Content Creators Present Firms and What Commenters
Express in Direct Threads***

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Abstract

On TikTok, sustainability claims and portrayed conduct appear in company posts, creator partnerships, news summaries, independent commentary, and personal testimony. Displayed platform counts show activity, but not what commenters evaluate. This thesis links 216 sustainability-related video–comment threads involving 40 S&P 500 firms with 13,235 direct comments. Context-aware coding and firm-grouped models distinguish substantive-comment yield, conditional firm criticism, and context-aware negative sentiment. Platform engagement rate showed no clear association with substantive-comment yield. Diagnostic framing accompanied higher yield; corporate or partnered source form accompanied lower yield than independent commentary or platform material; and critical or skeptical creator stance accompanied higher conditional firm criticism than supportive stance. Substantive-comment yield is the share of direct comments discussing sustainability; conditional firm criticism is opposition to the firm among issue-focused comments with a stable firm evaluation. The planned misaligned-versus-aligned comparison for conditional firm criticism was imprecise, and the source-adjusted comparison for context-aware negative sentiment was not significant after multiple-comparison adjustment. Most credibility challenges contained neither an explicit expectation nor action-oriented wording. An exploratory represented-relation comparison also showed a higher non-firm primary-target rate for misaligned than aligned videos. Linking each comment to its video and eligible base distinguishes activity from what commenters discuss, criticize, challenge, expect, propose, and evaluate. For organizations, the linked approach offers a structured way to examine how a firm’s sustainability positioning and portrayed conduct are evaluated in selected discourse, assess communication against its intended outcome, and turn broad criticism into a specific question for evidence review. These observational associations describe publicly visible evaluation in the selected threads.

Keywords: sustainability communication; TikTok; commenter evaluation; linked video–comment analysis; context-aware content analysis; corporate strategy

1. Introduction

A sustainability video involving a firm and posted on a social-media platform can attract thousands of “likes” and comments while prompting almost no discussion of sustainability. Its comments may instead focus on the product, the creator, a joke, or an unrelated detail.

Meanwhile, another thread may turn the same broad topic into visible evaluation of a firm: commenters examine a claim, question its evidence, or debate the conduct shown. This study therefore examines direct-comment threads to identify whether commenters discuss the sustainability issue, what they question, and whom they evaluate.

Companies often appeared in TikTok sustainability content through several routes: a firm could post for itself, work with a creator, feature in news coverage, or become the subject of independent commentary or personal testimony. TikTok allows users to search for content (TikTok, n.d.-a), follow accounts (TikTok, n.d.-b), and receive personalized recommendations (TikTok, n.d.-c), but its engagement counts such as views, likes, shares, saves, and comments record engagement rather than evaluation—whether there is approval or criticism. Research treats social-media engagement as multidimensional and dependent on platform and relationship (Brodie et al., 2013; Hollebeek et al., 2014; Voorveld et al., 2018); studies in political settings also show that morally charged or hostile material can attract interaction (Brady et al., 2017; Rathje et al., 2021).

TikTok is especially relevant to younger audiences. In a representative U.S. survey, 63% of adults aged 18–29 reported using TikTok, compared with 12% of adults aged 65 or older (Gottfried & Park, 2025). Research with 241 German Gen Z TikTok users also examined how sustainable-brand videos relate to attitudes and purchase intentions (Plötz et al., 2023). These patterns make TikTok a strategically relevant setting for studying sustainability communication on a platform with a strong younger-user presence.

Comments reveal what counts alone cannot: whether someone discusses the sustainability issue, criticizes the firm, reacts negatively to another actor or situation,

questions an identifiable claim, or states an expectation or action. These forms of response answer different questions. A negative sentence may concern someone other than the firm, and a credibility challenge may stop short of proposing a remedy. The study therefore records what each comment addresses, whom it evaluates, and which analytical questions it can answer.

Each comment is one expression by an observed participant in a thread; it does not speak for customers, affected stakeholders, or the wider public. Even so, linked comments can help organizations examine how a firm's sustainability positioning and portrayed conduct are evaluated in the selected discourse and identify a specific concern for review. They can show what question to examine next, while additional evidence is needed to judge how common, accurate, material, or actionable the concern is.

1.1 Research Problem and Gap

Prior research provides several analytical distinctions for examining sustainability communication and public response. Engagement studies separate different experiences and behaviors rather than treating all interaction as one stable quantity (Brodie et al., 2013; Hollebeek et al., 2014; Voorveld et al., 2018). Corporate social responsibility (CSR) communication studies examine source-related credibility conditions and differences between firm and stakeholder issue emphasis (Dunn & Harness, 2019; Gómez-Carrasco et al., 2021). Other work examines sustainability-message content and public interaction with those messages (Jha & Verma, 2024; Saxton et al., 2019). In experiments, participants who received conflicting information about a company's CSR statements and its behavior, results, or motives were more likely to perceive the company as hypocritical and respond negatively (Lewin & Warren, 2025; Wagner et al., 2009). Social-media sentiment makes affect visible, while organizational legitimacy remains a broader construct (Etter et al., 2018). Together, these strands of the literature explain why counts, claims, sources, reactions, and legitimacy should not be collapsed into one measure.

Several studies already connect social-media content with comments, but their analytical designs differ. Sallaku et al. (2025) combine manual interpretation of nine Shein-related TikTok videos with text-only VADER sentiment¹ across 19,816 associated comments; Chi et al. (2026) examine 131 public-relations packaging videos and 34,212 first- and second-level comments, then interpret a packaging-keyword subset of those comments in TikTok context; Beckmann et al. (2025) qualitatively read linked sustainability-influencer posts and comments; Duan et al. (2024) relate audiovisual video codes to patterns of who replies to whom rather than comment meaning; Pera and Aiello (2024) use transcript or description text and Sentence-BERT video–comment alignment²; and Liou et al. (2026) study CSR-related Reddit posts and comments that explicitly name S&P 500 firms, combining manual post coding, topic modeling, and SiEBERT sentiment³. These precedents define the comparison against which the narrower integration gap can be stated.

The remaining gap concerns how these elements are combined. In particular, this combination remains underdeveloped in a selected multi-firm TikTok corpus. It has four parts: creator-side coding of source form, creator stance, framing, claim visibility, represented conduct domain, and represented relation; context-preserving linkage to each video's direct-comment thread; separate eligible bases for the comment-side outcomes; and firm-grouped comparisons that check whether the required category combinations exist in the data. Each component has precedent. The contribution of this study lies in combining them so that the main findings can be traced from creator presentation, through commenter expression, to the eligible evidence base for each question.

¹ VADER is a rule-based sentiment tool designed for social-media text. It uses a sentiment dictionary and language rules to estimate whether a text is positive, neutral, or negative (Hutto & Gilbert, 2014).

² Sentence-BERT converts sentences into numerical representations that can be compared for semantic similarity. Pera and Aiello used it to measure how closely video text and audience comments were related in meaning (Reimers & Gurevych, 2019).

³ Topic modeling identifies recurring themes from word patterns across many texts. Liou et al. used Latent Dirichlet Allocation for this step. SiEBERT is a machine-learning model trained to classify English text as positive or negative (Blei et al., 2003; Hartmann et al., 2023).

1.2 Research Question and Linked Design

This thesis links those two sides in two overall research questions: How do content creators present companies' sustainability positioning and sustainability-related conduct in TikTok videos, and what forms of evaluation do commenters express in the direct comments beneath those videos?

Three guiding questions translate the research question into tasks that the data can answer:

1. How do content creators present the firm and sustainability issue, including the source, stance, framing, claim, and portrayed fit or conflict between positioning and conduct shown or described?
2. Which features of the videos are associated with more issue-focused discussion and more criticism of the firm?
3. What further distinctions emerge when context-aware negative sentiment, credibility challenges, explicit expectation and action-oriented wording, and the primary evaluative target are examined separately?

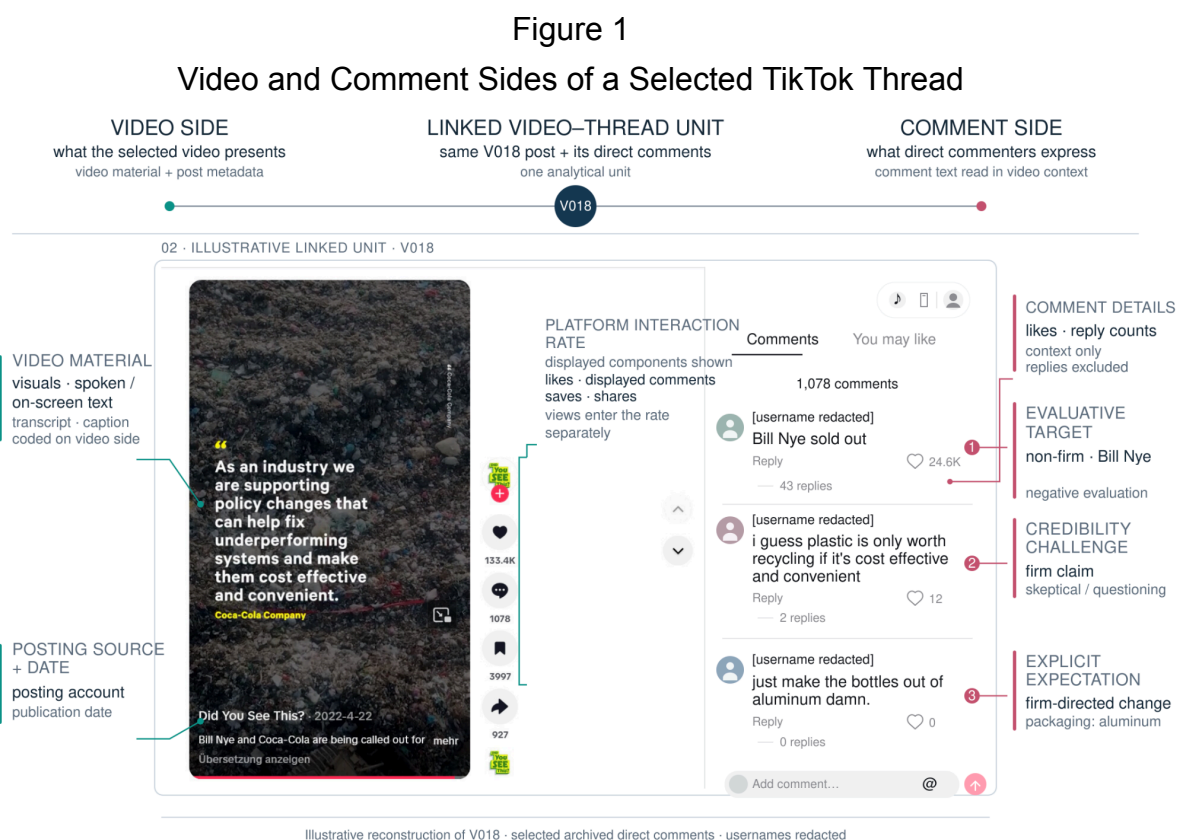
The three guiding questions have different jobs. The first is descriptive and concerns creator presentation. It is answered through video coding of source form, creator stance, framing, claim visibility, represented conduct domain, and whether the positioning–conduct relation is portrayed as aligned, misaligned, or other/ambiguous. Here “conduct” means what the video shows or describes the firm as doing. Assessing actual firm conduct would require external evidence.

The second guiding question is associational. It links coded video characteristics to two comment outcomes: substantive-comment yield—the share of direct comments that discuss the sustainability issue—and, among such comments with a stable firm evaluation, conditional firm criticism. In these regressions, the video characteristics are independent variables and the comment-thread measures are dependent variables.

The third guiding question separates additional forms of commenter evaluation. Context-aware negative sentiment and, exploratorily, the non-firm primary-target rate were modeled as outcomes. Credibility challenges, explicit expectations, and action-oriented wording were described using the comments able to answer each question. “Direct” or “top-level” identifies a comment’s structural attachment to a sampled video; relevance and target coding then determine which analytical questions that comment can answer.

The empirical unit links one selected video to its direct-comment thread. The archived retrieval ledger began with 5,424 query–video traces before consolidation, shortlisting, and contextual review selected 216 video–comment threads involving 40 firms and 13,235 direct comments. Comments were coded first; each outcome was then formed from its own numerator and eligible denominator within the thread. Firm-grouped models compare observed video characteristics with those thread outcomes. A targeted comparison with a published sentiment classifier contrasts labels based only on the written comment with the study’s context-aware sentiment labels in one narrow task, while 29 qualitative thread cases illustrate possible mechanisms, counterexamples, and difficult boundaries. Together, these components answer the research question at the level the data support: creator presentations and commenter evaluation in selected direct-comment threads.

Figure 1 illustrates the study’s linked video–comment design. It places one selected TikTok video beside selected direct comments from the same thread, making the two analytical sides visible: what the video presents and what commenters express.



Note. Illustrative reconstruction of video number 18 (V018) from the archived corpus, linking a video from the posting account “Did You See This?” (22 April 2022) to three selected direct comments from the same thread. Post-level likes, displayed comments, shares, and saves are components of the view-normalized platform engagement rate. Comment-level likes and reply counts are shown only as interface context; comment replies were excluded from coding. Usernames and profile images were removed, and the comments retain their archived wording after privacy review. The examples illustrate analytical distinctions. Author’s reconstruction from archived study materials.

1.3 Contribution and Main Findings

This study contributes to the literature by integrating sustainability communication, commenter evaluation, and corporate accountability. It connects video-coded source form, creator stance, framing, claim visibility, and the represented relation between a firm’s sustainability positioning and conduct shown or described with what commenters discuss, criticize, doubt, and request, and whom they evaluate. The academic value lies in showing how these questions connect without reducing them to one engagement or sentiment outcome. The methodological value lies in keeping video context and the eligible denominator visible for each question and grouping repeated observations by firm. Checks of category support,

field-specific recoding, a targeted comparison with a published sentiment classifier, documented reruns, and qualitative cases answer different validation questions. For organizations, the design connects two bounded uses: assessing communication against its intended outcome and turning broad criticism into a defined question for evidence review. By identifying what is challenged, whom or what is evaluated, and whether an expectation or proposed action is stated, it clarifies what to examine next. Application still requires organizational judgment and prospective testing.

Three associations remained clear after accounting for the multiple coefficient tests. Videos with diagnostic framing accompanied higher substantive-comment yield; corporate or partnered source form accompanied lower substantive-comment yield than independent commentary or platform material; and critical or skeptical creator stance accompanied more conditional firm criticism than supportive stance. The planned misaligned-versus-aligned comparison pointed toward more conditional firm criticism but remained inconclusive, while the higher estimate for context-aware negative sentiment did not meet the significance threshold after adjustment for multiple comparisons. The near-null platform-engagement–yield relation and the credibility-challenge, explicit-expectation, and action-oriented-wording profiles are descriptive findings; the represented-relation comparison for non-firm primary-target rate remains exploratory. This evidence hierarchy shows why engagement and sentiment numbers answer only part of what the threads express.

Eligible bases keep each conclusion traceable: a substantive comment may evaluate a regulator, context-aware negative sentiment may concern an event or price, and a credibility challenge may contain no explicit response request.

2. Theoretical Framework

This chapter develops four distinctions that guide the coding, eligible bases, and interpretation: claims versus portrayed conduct; communication setting versus audience response; visible activity versus forms of evaluation; and primary evaluative target versus

wider accountability roles. Each distinction also states the boundary of what the evidence can support.

2.1 Claims, Conduct, and Credibility

A TikTok video offers a selective portrayal built from speech, captions, on-screen text, images, events, claims, and references to different actors. Viewers and commenters may interpret that partial account differently. Sensemaking concerns how people make retrospective sense of ambiguous situations (Weick, 1995), and Basu and Palazzo (2008) apply that lens to managerial and organizational interpretations of corporate social responsibility. Here, sensemaking helps explain why commenters may interpret the same portrayal differently. The study therefore examines what they support, question, criticize, or request in the selected threads.

Legitimacy concerns whether an organization, claim, or conduct is judged appropriate within a wider system of norms and expectations. Research distinguishes who makes the judgment, what object is judged, and whether the basis is pragmatic, moral, or cognitive (Deephouse et al., 2017; Suchman, 1995). It also separates legitimacy from reputation and status (Bitektine, 2011). A comment that questions a sustainability claim can therefore participate in a legitimacy process without measuring the firm's overall legitimacy. Individual judgments and collective legitimacy operate at different levels (Bitektine & Haack, 2015; Tost, 2011). This matters here because the study observes different judgments in selected threads, rather than a population-level assessment (Etter et al., 2018, 2019).

A visible claim gives commenters something specific to examine. They may question its evidence, scope, wording, omissions, or consistency with conduct shown or described. The study treats this possibility as claim contestability and codes a credibility challenge only when a comment explicitly questions an identifiable proposition's truthfulness, adequacy, consistency, or evidentiary support. Greenwashing research distinguishes forms and drivers of misleading environmental communication (de Freitas Netto et al., 2020; Delmas & Burbano,

2011; Lyon & Montgomery, 2015). The present study uses that literature to interpret credibility challenges but does not classify the sampled videos as greenwashing. An external allegation may contribute to a legitimacy process while remaining an expressed challenge rather than proof of deception (Seele & Gatti, 2017).

The video code also records how the creator portrays the relationship between a firm's sustainability positioning and the conduct shown or described. This variable is called represented relation. It is aligned when the creator presents the two as consistent, misaligned when the creator presents a conflict, and other or ambiguous when neither interpretation is clear. Cho et al. (2015) distinguish between what firms say about sustainability and what they do. Lewin and Warren (2025) and Wagner et al. (2009) found that conflicts between CSR statements and information about company conduct, results, or motives could increase perceived hypocrisy and negative responses. This literature supports comparing videos that portray sustainability positioning and conduct as aligned or misaligned. Together, the three fields separate the claim presented, the creator's portrayal of the relationship between positioning and conduct, and any credibility challenge expressed by commenters. Assessing factual accuracy requires external evidence.

2.2 Source, Stance, and Framing

The same sustainability issue can reach TikTok through a company post, a creator partnership, a news summary, independent commentary, or personal testimony. This source setting may shape how the issue and firm enter the visible public discourse. The study therefore records source form, whether the creator's stance toward the firm is supportive, critical/skeptical, or neutral/mixed/unclear, and whether the video frames a problem, a response, or a call to engage or act. Together, these variables describe the communication setting attached to each thread.

Framing makes selected aspects of an issue more prominent (Entman, 1993). A diagnostic frame explains a problem, its cause, or responsibility; a prognostic frame proposes a solution or response; and a motivational frame encourages engagement or action. Benford

and Snow (2000) distinguish these tasks and show why their audience connection depends on context and credibility. In this study, a diagnostic frame may give commenters a clearer object to discuss. This is a plausible interpretation of the association rather than evidence that the frame caused substantive commenting.

People may also evaluate the communicator and the persuasive attempt.

Persuasion-knowledge theory proposes that people draw on what they know about persuasive agents and tactics when interpreting communication (Friestad & Wright, 1994). Access to persuasion knowledge has often been linked to skepticism, yet Isaac and Grayson (2017) report conditions in which access instead bolstered credibility when the persuasive agent used a credible tactic. CSR studies likewise report source-related differences in skepticism and issue emphasis in their own settings (Dunn & Harness, 2019; Gómez-Carrasco et al., 2021). These findings motivate examining source form while leaving the direction of response open to the data.

Creator stance records the direction of the creator's portrayal toward the firm. A critical creator may foreground an alleged inconsistency, while a supportive creator may foreground an initiative or evidence offered by the firm. Thread responses may also vary with the issue, prior firm history, creator, audience, venue, and exposure. Studies of Instagram visibility, social-listening venue choice, and observational-network confounding show how selection processes can shape who and what becomes visible (Cotter, 2019; Schweidel & Moe, 2014; Shalizi & Thomas, 2011). The method therefore treats source, stance, and framing as observed features associated with thread outcomes, not as isolated causal treatments.

2.3 Platform Engagement Rate and Forms of Evaluation

Platform engagement rate and evaluation answer different questions. A single thread can show substantial platform activity and several forms of evaluation. The study therefore asks how much visible activity formed around a video, what share of direct comments discussed the sustainability issue, how firm-directed comments expressed conditional firm criticism,

whether comments expressed context-aware negative sentiment, and whether commenters challenged a claim or stated an explicit expectation or action-oriented wording. Each question receives its own observable and eligible base.

To describe visible platform activity, the study uses the view-normalized platform engagement rate: likes, the platform-displayed comment count, shares, and saves divided by views. Here, engagement denotes the visible platform actions captured by these counts; cognitive or emotional engagement lies outside the measure. It supplies activity context; comment coding determines what that activity contains. Voorveld et al. (2018) show that engagement experiences differ across platforms. Cheng and Li (2024) find in TikTok news videos that likes, comments, and shares were associated with different message features, while political-network studies show that activity can accompany moral-emotional or antagonistic language (Brady et al., 2017; Rathje et al., 2021). Together, these studies show that visible engagement counts can carry different meanings. Separately, this study measures substantive-comment yield from the coded comment content.

The study defines substantive-comment yield as the share of direct comments that discuss the sustainability issue. Its numerator is Tier-A issue-focused comments, and its denominator is all retrieved direct comments eligible for tiering. If 20 of 100 comments discuss sustainability, the yield is 20%; comments about a joke or product remain in the denominator. Conditional firm criticism asks a narrower question: among Tier-A comments that clearly evaluate the firm, what share is coded as opposing the firm? The smaller denominator preserves the meaning of that second question.

Context-aware negative sentiment asks whether a comment expresses negative affect, regardless of whether its object is the firm, a regulator, an event, a price, a technology, a creator, or a wider situation. It is related to sentiment, but the video context is retained because linguistic context can change a phrase's polarity, and implicit or discourse-level sentiment can be missed when relevant context is absent (Villarroel Ordenes et al., 2017; Wilson et al.,

2005). Context-aware negative sentiment is therefore distinct from conditional firm criticism and from blame, which also involves causal information, intentionality, and norm violation (Malle et al., 2014). Section 3.4 compares the context-aware labels with labels assigned by a published sentiment classifier that received only the written comment.

Credibility challenges, explicit expectations, and action-oriented wording provide information beyond whether a comment is positive or negative. Consider a hypothetical exchange: “That number does not support the claim” challenges the evidence; “The company should publish the full data” states an explicit expectation; and “Report it to the regulator” proposes an action. The coding records these expressions separately. Together, the fields distinguish comments that identify a problem with a claim from comments that also state an expected response or propose a next step.

2.4 Evaluative Targets and Accountability

Comments can direct evaluation and requested action toward different actors. A comment may criticize a firm while asking a regulator to change a rule. The study therefore distinguishes who or what receives the main evaluation, who is portrayed as contributing to the problem, who could help solve it, who is expected to respond, and who receives a requested action. The same negative wording raises a different evidence question when it concerns a firm, institution, consumer, creator, or wider system.

Accountability, in its fuller sense, connects an actor with a forum that can request information, question and judge an account, and possibly impose consequences (Bovens, 2007). The study observes only the parts of that relationship expressed in comment language. Corpus-wide coding records the primary evaluative target. In addition, 29 purposively selected qualitative thread cases examine the actor portrayed as contributing to the problem, able to offer a solution, expected to respond, or addressed by an action. These cases illustrate how the roles can separate; they do not estimate how often each role occurs in the corpus.

When many actors contribute to a problem, causal contribution and responsibility can be difficult to allocate. Thompson (1980) and van de Poel et al. (2012) examine this many-hands problem, while Young (2006) develops a shared, forward-looking social-connection model. Read together as an author synthesis, these perspectives explain why the study keeps actor roles open for investigation rather than assuming that a comment allocates responsibility correctly. Implicit references can also remain unresolved. In a hypothetical comment such as "They should regulate this," the expected actor is coded only when the surrounding video and comment context identify who "they" refers to. A primary evaluative target therefore records only who receives the main evaluative language; causal, moral, and legal responsibility require external evidence and the relevant normative or legal standard.

Keeping these roles separate prevents an observation from becoming a recommendation too quickly. If a comment criticizes a firm but asks a regulator to act, the firm is the evaluative target and the regulator is the action recipient. If it criticizes a wider system while asking the firm to disclose evidence, those roles reverse. Coding only negative tone would miss both structures, while coding one primary evaluative target preserves only part of them. For selected threads, the qualitative case layer therefore records the actor portrayed as contributing to the problem, able to offer a solution, expected to respond, or addressed by an action; the corpus-wide target field remains focused on the main actor or object evaluated.

Table 1 connects §§2.1–2.4 to their questions, observables, and interpretation boundaries; Section 3 defines each measure and eligible base.

Table 1
Four Analytical Distinctions and Their Use in the Study

Analytical distinction and subsection	Main analytical question	Project observables and later use	Interpretation boundary
§2.1 Claims, Conduct, and Credibility	What claim and portrayed conduct appear, and do commenters challenge the claim?	Video: claim visibility, conduct domain, and represented relation. Comments: credibility challenge. Used in §§4.3–4.5.	Portrayals and expressed challenges require external evidence to assess truth, actual conduct, intent, or greenwashing.
§2.2 Source, Stance, and Framing	Who presents the issue, with what stance and frame?	Video: source form, creator stance, and diagnostic, prognostic, and motivational framing. Used as descriptive features and predictors.	These fields describe the observed setting. Causal claims about persuasion, trust, or response require another design.
§2.3 Platform Engagement Rate and Forms of Evaluation	How much activity is visible, and what forms of evaluation appear in the thread?	Activity: platform engagement rate. Comments: substantive-comment yield, conditional firm criticism, context-aware negative sentiment, credibility challenge, explicit expectation, and action-oriented wording. Used in §§4.2–4.5.	Each measure has its own meaning and eligible base. Population opinion and importance require other evidence.
§2.4 Evaluative Targets and Accountability	Who or what is evaluated, and which actors are linked to the problem, solution, expectation, or action?	Comments: primary evaluative target and additional actor roles in qualitative cases. Used in §4.5 and Appendix E.	Target identifies the evaluated actor. Factual, causal, moral, or legal responsibility requires separate evidence; cases are illustrative.

Note. Video fields describe selected portrayals; comment fields describe expressions in retrieved direct comments. Chapter 3 defines each measure and eligible base. Author's conceptualization based on the literature synthesized in §§2.1–2.4.

3. Methodology

3.1 Study Design and Units

The study links video-level coding of creator presentation with context-aware coding of each direct comment. Video fields capture source form, creator stance, framing, claim visibility, represented relation, conduct domain, and contextual material; comment fields capture substantive discussion, conditional firm criticism, context-aware negative sentiment, credibility challenge, explicit expectation, action-oriented wording, and primary evaluative target. For each thread outcome, the analysis retained the number of comments meeting its criterion and the number eligible to answer it. Firm-grouped models relate video characteristics to these thread outcomes. A targeted comparison with a published sentiment classifier and purposively selected cases provide measurement and interpretive checks. Figure 1 illustrates the linked unit.

The evidence covers social media videos and written comments in the retrieved threads. Silent viewers and individual exposure paths cannot be recorded. Platform-trace

research likewise requires explicit limits on representativeness and inference (Olteanu et al., 2019; Tufekci, 2014). The study describes observable creator presentations, commenter expressions, and associations in this corpus.

3.2 Retrieval, Screening, and Final Sample

Discovery began with a screening of S&P 500 firms. The S&P 500 provided a predefined cross-sector screening frame of major U.S. public companies that receive sustained attention from investors and other stakeholders, including attention to corporate strategy and sustainability-related targets. The frame also allowed the same procedure to be applied systematically across firms. The screening file represented 501 distinct search aliases. Each alias was assessed with three predefined queries—[alias] sustainability, [alias] environment, and [alias] greenwashing—creating 1,503 distinct alias–query combinations for evaluation. To filter for useful keyword searches, the researcher removed searches with insufficient relevant-video yield. A score of 2 or 3 on any of a firm’s three queries selected that firm for metadata retrieval. Within a selected firm, queries scored 1–3 proceeded, while score-0 queries were excluded. This rule selected 43 firms and produced 110 search routes.

Video metadata were retrieved with the Clockworks TikTok Scraper on the Apify platform.⁴ The record and Appendix J.1 preserve the dates, search strings, documented inputs, settings, and archived exports. The retrieval comprised 111 executed query strings and produced 5,424 query–video traces representing 5,210 exact Uniform Resource Locators (URLs). A trace is one URL returned through one exact query, so the same URL can appear more than once before consolidation. The researcher retained every query route and then removed duplicate URLs. Appendix J.1 preserves the retrieval settings, score distribution, overlap counts, exceptions, and final-firm restriction. Appendix Tables J3 and J4 compare how the selected videos and outcomes differed across the three search terms.

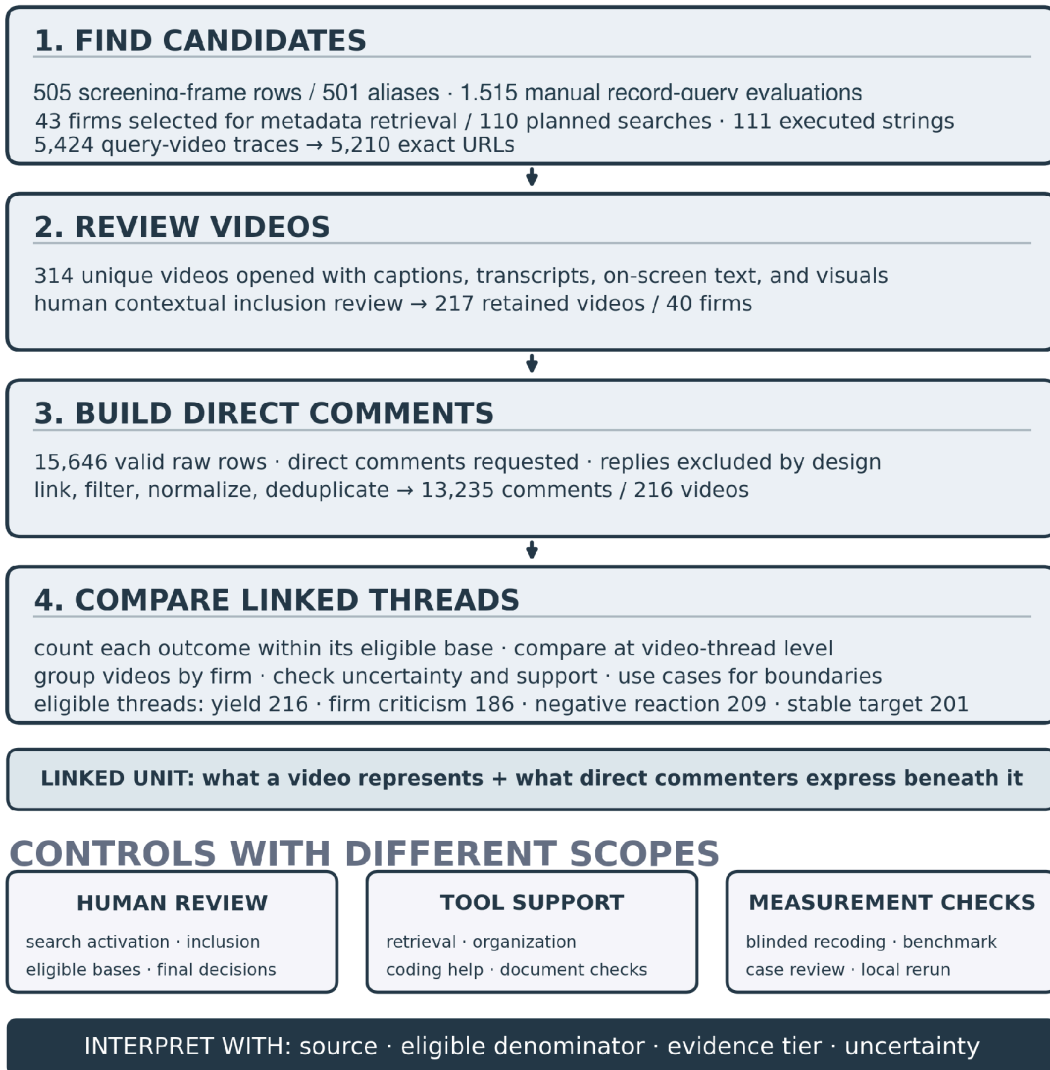
⁴In this study, Apify was the online platform used to run two third-party Clockworks Actors and export their dataset outputs (Apify, n.d.-a, n.d.-b; Clockworks, n.d.-a, n.d.-b). One Actor retrieved public video information for the specified searches; the other retrieved direct comments for selected videos. These tools were not TikTok’s official Research API (application programming interface). An API is a structured way for software systems to exchange data. A Uniform Resource Locator (URL) is a web address. When the same video URL appeared in several searches, the study retained one video record while preserving every search route through which it was found.

Generative artificial intelligence (AI) supported the prioritization of video candidates for researcher review; the researcher made every final inclusion decision. After URL consolidation, all 314 shortlisted video candidates received researcher review using the available transcripts or captions, on-screen text, visuals, and contextual notes. The review assessed whether each candidate had an identifiable focal firm, sustainability positioning or conduct shown or described, enough multimodal evidence for interpretation, and an analyzable link between video and thread. This review retained 217 video records; 216 had usable direct-comment threads and formed the main linked analytical corpus across 40 firms. Available transcripts, captions, on-screen text, and visual or contextual notes documented the creator material used for portrayal coding. Appendix F describes the scope of generative-AI assistance and researcher oversight.

After video selection, the Clockworks TikTok Comments Scraper requested up to 200 direct (top-level) comments and no replies per video; collected threads contained 3–200 comments. The 200-comment limit was a collection cap. Figure 2 traces the changing units from distinct aliases and search evaluations to retained videos, finalized comments, and outcome-specific thread bases, while its lower bands distinguish researcher review, tool support, and measurement checks.

Figure 2
From Search Results to the Final Video–Comment Sample

FOUR LINKED STAGES



Note. The main diagram reports the deduplicated basis of 501 distinct aliases and 1,503 distinct alias–query evaluations. Appendix J retains the historical 505-row ledger and reconciles it with the distinct-alias basis. The diagram then changes units through metadata retrieval, reviewed videos, direct comments, and outcome-specific eligible video threads. Eligible thread counts are 216 for substantive-comment yield, 186 for conditional firm criticism, 209 for context-aware negative sentiment, and 201 for stable primary evaluative target. The control blocks have different scopes. Author’s analysis of the documented project process record and final analytical dataset.

3.3 Variable Coding and Eligible Bases

Written operational rules structured the video coding and the interpretation of linked comments in context. Krippendorff (2018) supports explicit recording and coding instructions in context; the categories below are this project's instrument. Creator presentation is the umbrella term for what the creator showed, said, or described about the firm and sustainability issue; it was not a separate regression variable. The video-level variables were:

- **Represented relation.** Whether the creator portrayed the firm's sustainability positioning and portrayed conduct as aligned, misaligned, or other/ambiguous.
- **Source form.** Whether the video was corporate/partnered material, independent commentary/platform material, a news summary, or personal testimony.
- **Creator stance.** Whether the creator's position toward the firm was supportive, critical/skeptical, or neutral/mixed/unclear.
- **Claim visibility.** Whether the video contained an explicit sustainability proposition, an attributed/reputational proposition, or no clear proposition.
- **Sustainability centrality.** Whether sustainability was the video's main subject or part of the surrounding context.
- **Represented conduct domain.** Whether the conduct shown or described concerned strategic/governance/finance, operational/environmental, communication/claim, product/service, or other/unclear.
- **Diagnostic, prognostic, and motivational framing.** Three separate present/absent indicators. Diagnostic framing explained a problem, cause, or responsibility; prognostic framing proposed a solution or response; and motivational framing encouraged engagement or action. More than one could be present in the same video.

These codes describe observable communication; actual conduct, affiliation, sponsorship, intent, deception, and belief require separate evidence.

Generative AI applied the final researcher-defined prompt and codebook to the 13,235 direct comments together with the supplied comment text and archived video–thread context. The researcher developed and approved the categories, decision rules, and analytical boundaries, reviewed the task-specific validation results, and decided how the coded fields entered the analysis and interpretation. Appendix F documents this assistance and its limits.

Each direct comment then entered one of four relevance tiers. Tier A contained a stable, context-interpretable proposition, evaluation, testimony, factual assertion, counterclaim, expectation, or reported or recommended action about the focal firm or a materially connected actor or issue. Tier B retained comments clearly linked to the video but expressed mainly through affect, agreement, surprise, sympathy, interest, or information seeking without a stable substantive proposition. Tier C covered comments plausibly connected to the video whose meaning or target remained unresolved after review. Tier X excluded comments with no usable relationship to the research question, including unrelated product or customer-service chatter, spam, duplicate text, and tag- or emoji-only content. The totals were 7,302 Tier A, 1,876 Tier B, 97 Tier C, and 3,960 Tier X, summing to 13,235. The tiers route comments to the questions they can answer; they do not rank comments by correctness or value. Substantive-comment yield uses Tier A rather than Tier A/B because it asks whether a comment contributes a stable, interpretable sustainability-related proposition or evaluation. Tier B remains eligible for sentiment and action-oriented wording, for which stable propositional content is not required.

The eligible denominator changes with the question. If 12 of 20 retrieved comments in a thread meet Tier A, substantive-comment yield is 12/20. If only five of those 12 clearly evaluate the firm and four are coded against it, conditional firm criticism is 4/5. The smaller second denominator preserves the meaning of the criticism question by using only comments capable of answering it.

- TikTok's official Research API codebook identifies like, comment, share, view, and favorites counts as video-level metadata (TikTok, n.d.-d). This study's platform engagement rate used the corresponding archived video metadata fields: (likes + video-level comment count + shares + saves) / views. The comment-count term is the video's metadata snapshot, not the 13,235-comment research corpus.
- **Substantive-comment yield** is the share of direct comments that discuss the sustainability issue: 7,302 Tier-A comments among 13,235 direct comments across 216 videos from 40 firms.
- **Conditional firm criticism** records opposition among issue-focused comments with a stable firm target: 1,893 against-firm comments among 3,260 eligible comments across 186 videos from 38 firms.
- **Context-aware negative sentiment** records negative affect wherever its object lies. The eligible base was 8,898 Tier-A/B comments with a valid three-class context-linked sentiment label; 280 Tier-A/B comments without one were outside this denominator. The outcome contained 4,105 negative classifications across 209 videos from 38 firms.
- **Non-firm primary-target rate** records the share of stable Tier-A primary evaluative targets assigned to an institution or system, a consumer, or the creator. Its pool count was 3,146 non-firm targets among 6,406 stable targets across 201 videos from 38 firms. A further 890 other-or-unclear assignments and six unstable assignments remained outside the denominator.

Credibility challenge, explicit expectation, and action-oriented wording have related but different bases. At comment level, explicit mismatch—a separate code from video-level represented-relation category—recorded whether a Tier-A comment itself contrasted a firm's sustainability positioning with conduct shown or described. Among 7,302 Tier-A comments, 557 were credibility challenges (7.63%). Credibility challenge and explicit expectation were usually Tier-A fields because both required a stable substantive proposition. Action-oriented

wording admitted Tier A and Tier B because a context-linked reaction could report or recommend behavior without making a stable evaluation. Explicit expectation appeared in 1,389/7,302 Tier-A comments; action-oriented wording appeared in 1,845/9,178 Tier-A/B comments.

The code “neither” marks comments that state neither an expectation nor an action; unstated preferences and later behavior lie outside the coding. Comments with missing or unstable labels remain outside the relevant outcome. Across the coding system, each question uses the comments able to answer it, and ambiguous records remain unassigned.

3.4 Validation and Qualitative Cases

Validation addressed four questions. First, how consistently did the same AI system reproduce selected comment fields in two independently executed predictor-blinded runs? Second, how closely did an independent recoding pass for selected video/context fields agree with the original assignments? Third, how did labels from a published sentiment classifier that received only the written comment compare with the researcher-reviewed context-aware labels in a targeted set for this study? Fourth, did rerunning the saved classifier reproduce the same labels and comparison statistics? Context can alter the polarity of a phrase (Wilson et al., 2005), and surrounding language or video features can improve sentiment classification in user-generated-video settings (Poria et al., 2017). In this study, a short comment could also depend on an implied target or linked video context that was unavailable to the classifier.

For the field-specific reproducibility check, the same configured AI system recoded the same 360 comments from 126 video groups in two independently executed, predictor-blinded runs, meaning that the recoding input did not include the stored video-level predictor labels later used in the regressions. The set combined 240 probability-sampled corpus comments across all six coding batches and four tiers with 120 deliberately targeted comments. The probability component covered ordinary corpus cases, while the targeted component stressed difficult or substantively important categories; the combined

360-comment set was therefore not a simple probability sample of the full corpus. Agreement ranged from 77.8% (Cohen's $\kappa = .738$) for primary evaluative target, the least stable field, to 89.4% ($\kappa = .827$) for direction toward the firm. Appendix Table G2 reports agreement for the recoded fields: tier, direction toward the firm, sentiment, legitimacy, credibility challenge, comment-level explicit mismatch, explicit expectation, action-oriented wording, sarcasm, and primary evaluative target. Inter-run disagreements were diagnostic and did not automatically replace the final project coding. These coefficients describe run-to-run consistency of the configured AI system for the selected coding fields, not coding accuracy or agreement between independent human coders (Krippendorff, 2004). Agreement of each pass with the stored final coding was lower for some fields, particularly primary evaluative target; the check therefore does not validate the stored labels, and the primary-target model remains exploratory.

A separate video-field check, by the researcher, compared the original assignments for six secondary video/context fields with an independent recoding pass on 72 selected videos. Agreement ranged from 76.4% to 91.7%, with Cohen's κ from .530 to .826. Stakeholder burden was checked separately as a binary field and had 86.1% agreement; the retained record does not preserve a κ for that check. Appendix Table G4 reports the fields and their limited descriptive use.

Because a short comment can depend on its linked video, thread, implied target, or tone, the study compared the researcher-reviewed context-aware sentiment labels with labels from a published sentiment classifier. The model was Cardiff NLP's *twitter-roberta-base-sentiment-latest*, a RoBERTa-based classifier trained on Twitter data and fine-tuned to label text as negative, neutral, or positive (Barbieri et al., 2020; Cardiff NLP, 2022; Liu et al., 2019; Loureiro et al., 2022). It received only the written text of 222 eligible comments, without the linked video or thread. It served as an external benchmark for this narrow task, not as ground truth or as a general evaluation of RoBERTa.

Of the 128 comments labeled negative after context-aware review, the classifier also labeled 73 negative, giving negative recall of 57%. Negative precision was 88%: 73 of the 83 comments labeled negative by the classifier were also labeled negative after context-aware review. The largest observed difference concerned comments labeled negative with video or thread context but neutral when only the written comment was supplied. Missing context may contribute, especially for implied targets, tone, or sarcasm, but the comparison also used a different labeling procedure.

The comparison shows that the two label sets were not interchangeable for this targeted task, especially for negative labels. Because both the information available and the labeling procedure differed, it does not isolate the effect of context or establish either label set as ground truth. It instead tests whether an isolated-text sentiment benchmark can substitute for the study's context-aware coding; in this set, it could not. A documented local CPU rerun reproduced all 13,235 classifier labels and the reported statistics for the 222-comment comparison. Section 4.6 reports the comparison, and Appendix G records the confusion matrix, implementation, and limits. Classifier labels therefore require task-specific validation, and context-aware coding also requires documented reproducibility checks (Grimmer & Stewart, 2013; Plank, 2022).

The qualitative analysis examined 29 purposively selected cases from 29 videos, 27 firms, and 501 selected comments. The number reflects documented interpretive roles—contrasting outcome patterns, actor roles, and difficult boundaries—rather than statistical representativeness or saturation. The cases help interpret the quantitative patterns and identify places where the codes support only a limited interpretation. The researcher reviewed all 29 final case interpretations; Appendix E reports the separate case-review outcomes and notes six additional human confirmations. Appendix E also gives the full thread counts for the four cases discussed in Chapter 4.

3.5 Statistical Analysis

The creator-presentation side of the research question was answered descriptively through the coded video characteristics. In the regressions, these video characteristics served as independent variables, and the comment-thread outcome shares served as dependent variables.

The analysis used two statistical approaches. Section 4.2 reports one Spearman rank correlation between platform engagement rate and substantive-comment yield across 216 videos. A correlation is symmetric, so neither measure was formally designated as the dependent or independent variable. It tests whether the two video-level measures tend to move together; reversing the axes therefore does not change the correlation coefficient, p value, confidence interval, or interpretation.

The final analysis contained 14 video-level beta-binomial regressions. Six use substantive-comment yield as the dependent variable, four use conditional firm criticism, and three examine context-aware negative sentiment: two use the negative outcome and one checks a negative-or-mixed reaction outcome definition. One exploratory model uses the non-firm primary-target rate. Depending on the model, the independent variables are represented relation, source form, framing, claim visibility, creator stance, or represented conduct domain. Sections 4.3–4.5 report the main results, while Appendix Table H4 maps the complete model set.

These beta-binomial regressions are generalized linear mixed models (GLMMs). For each eligible video thread, the model retained the number of comments meeting the outcome and the total number able to answer that outcome question. The beta-binomial form allowed more variation between video threads than a simple binomial model (Williams, 1982), while a firm random intercept accounted for multiple videos involving the same firm. The models were estimated in R with the glmmTMB package (Brooks et al., 2017).

All 14 regressions included the same five control variables: sustainability centrality, creator follower count, video age, video duration, and an indicator for missing duration.

Follower count, age, and duration were log-transformed and standardized before modeling. These controls account for measured differences in issue centrality, follower count, time online, and video length.

Before model selection, the study checked the observed combinations of represented relation with source form and creator stance. Missing or sparse combinations prevented reliable larger combined regression models. Separate outcome models therefore matched the comparisons supported by the observed data.

The study reports odds ratios (ORs) with 95% confidence intervals (CIs). Likelihood-ratio tests assessed each tested video characteristic as a whole. Because the regressions contain many coefficient comparisons, the Benjamini–Hochberg procedure adjusted 126 fixed-effect *p* values to reduce risk of false-positive findings across the set (Benjamini & Hochberg, 1995). Planned primary comparisons are reported with their coefficient *p* values; additional comparisons are reported with BH-adjusted *q* values, for which $q < .05$ indicates statistical significance after adjustment. Because the videos were observed rather than randomly assigned to source, stance, framing, claim, or represented-relation categories, the models estimate associations rather than causal effects (D’Amour et al., 2021; Stuart, 2010). Appendices D, H, and I report the detailed checks and sensitivity analyses; Appendix Table G3 distinguishes the original analysis, later residual diagnostic, and targeted verification environments.

The study analyzed publicly accessible videos and direct comments without contacting users. Reader-facing examples were generally paraphrased; Figure 1 retains three short archived comments with usernames and profile images removed. Appendix F documents the ethics, privacy, data-access, generative-AI, and researcher-oversight procedures (Franzke et al., 2020).

4. Results

Results proceed from sample composition and the engagement–yield correlation to regression, descriptive, qualitative, and diagnostic findings. Section 4.2 reports the sole Spearman correlation; §§4.3–4.5 report selected results from the 14 beta-binomial regressions, and §4.5 distinguishes descriptive profiles from the exploratory non-firm-target model.

4.1 Sample and Outcome Bases

The main analytical corpus comprised 216 linked video–comment threads involving 40 firms and 13,235 direct comments. Of those comments, 7,302 were Tier-A issue-focused comments. Table 2 reports the outcome-specific eligible bases; Appendix Table H3 retains the full video-only composition and descriptive record.

The full selected-video record contained 217 videos. Independent commentary or platform material was most common (103, 47.5%), followed by corporate or partnered material (50, 23.0%), news summaries (35, 16.1%), and personal testimony (29, 13.4%). Critical or skeptical stances (94, 43.3%) slightly exceeded supportive stances (84, 38.7%), and diagnostic framing appeared in 172 videos (79.3%). Represented relation was aligned in 91 videos (41.9%), other or ambiguous in 85 (39.2%), and misaligned in 41 (18.9%).

Table 2
Comment-Thread Outcomes and Eligible Bases

Outcome	Combined outcome / eligible total	Eligible videos / firms	Mean; median across eligible videos
Substantive-comment yield	7,302 / 13,235	216 / 40	55.8%; 60.4%
Conditional firm criticism	1,893 / 3,260	186 / 38	62.0%; 66.7%
Context-aware negative sentiment	4,105 / 8,898	209 / 38	43.6%; 43.8%
Non-firm primary-target rate	3,146 / 6,406	201 / 38	44.9%; 50.0%

Note. Each row uses its outcome-specific eligible base. Pooled numerators and denominators combine eligible comments across threads; means and medians are unweighted across eligible videos. These quantities answer different descriptive questions; neither estimates platform prevalence. Author’s analysis of the final analytical dataset.

Table 2 gives both pooled and unweighted video-level summaries because thread lengths differ. The regressions retain each video's outcome count and eligible total while accounting for repeated firms.

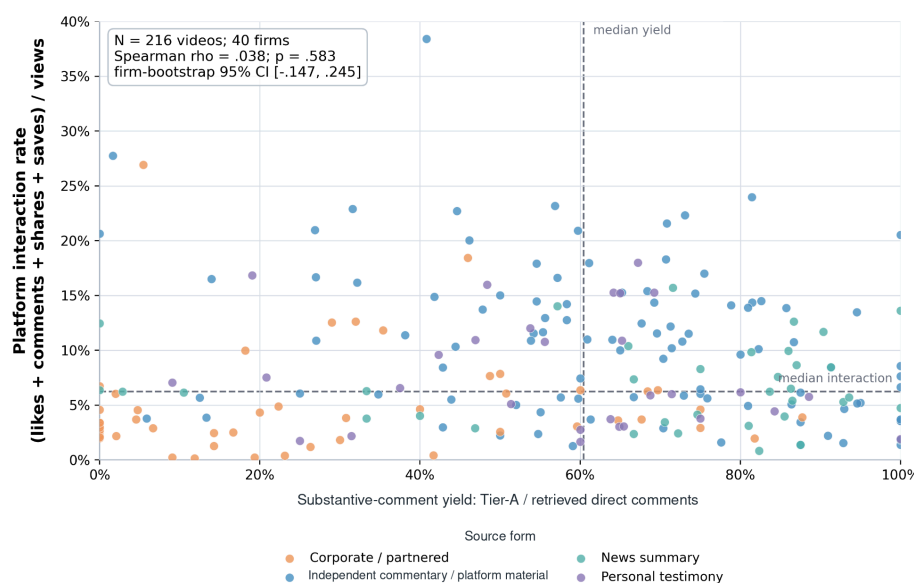
Eligibility also differs by question: comments may discuss sustainability without evaluating the firm or may be negative toward another actor. Such comments remain in broader outcomes but are excluded from narrower outcomes requiring a stable firm evaluation, valid sentiment label, or stable target. Appendix Table H3 provides the full profile.

4.2 Substantive-Comment Yield and Platform Engagement Rate

Substantive-comment yield and platform engagement rate, the study's measure of visible activity, showed almost no monotonic relationship. Across 216 videos from 40 firms, Spearman's ρ was .0376 ($p = .583$; 2,000-draw firm-bootstrap 95% confidence interval $[-.147, .245]$). This provides no clear evidence that videos with a higher share of issue-focused comments also had more visible activity, although the interval remains compatible with modest relationships in either direction.

The opposite quadrants show what that near-zero association looks like. Fifty-two videos with above-median platform engagement rate and below-median substantive-comment yield contained 1,694 Tier-A comments among 4,451 retrieved comments. Another 52 videos with below-median platform engagement rate and above-median substantive-comment yield, spanning 25 firms, contained 2,042 Tier-A comments among 2,598 retrieved comments. These sample-median groups describe contrasting thread profiles. Figure 3 displays the video-level comparison.

Figure 3

Substantive-Comment Yield and Platform Engagement Rate

Note. The scatterplot covers 216 videos from 40 firms ($\rho = .0376$, $p = .583$; 2,000-draw firm-bootstrap 95% CI $[-.147, .245]$). Substantive-comment yield is shown on the x-axis and platform engagement rate on the y-axis. Spearman correlation is symmetric, so this reading order does not create a dependent–independent direction. Dashed lines mark the sample medians; colors indicate source form. Source: Author’s analysis of the final analytical dataset.

Appendix Table A1 reports the four-quadrant video profile, and Appendix Table J2 compares pooled and unweighted activity summaries. The next analysis asks which video features accompanied greater issue-focused concentration.

4.3 Video Features Associated with Issue-Focused Discussion

In this section, substantive-comment yield is the dependent variable. The independent variables are coded video characteristics: represented relation, source form, framing, and claim visibility, tested in separate regression models.

Videos that explained a problem had threads with more issue-focused discussion. Compared with videos without a diagnostic frame, their threads had higher modeled odds of a Tier-A comment (odds ratio (OR) = 1.908, 95% confidence interval (CI) $[1.222, 2.978]$, $p = .00446$, $q = .02445$; overall framing test $p = .03837$). The estimate uses 7,302 Tier-A comments among 13,235 retrieved comments across 216 videos and 40 firms. In this corpus, diagnostic framing was the clearest positive video-level correlate of substantive-comment

yield. In the same source-adjusted framing model, prognostic and motivational framing showed no clear independent differences (prognostic OR = 1.074, $p = .706$; motivational OR = .845, $p = .374$).

The source-form contrast was even larger. Corporate or partnered videos had lower modeled odds of a Tier-A comment than independent commentary or platform material (OR = .298, 95% CI [.188, .471], $p = 2.32 \times 10^{-7}$, $q = 2.93 \times 10^{-5}$; overall source-form test $p = 1.71 \times 10^{-8}$). The corresponding pooled counts were 877/2,724 and 3,260/5,509. The category describes the observed source form; possible undisclosed connections were outside the coding.

Misalignment initially accompanied higher substantive-comment yield than alignment (OR = 1.896, 95% CI [1.178, 3.050], $p = .00839$; overall represented-relation test $p = .00108$). After source form was added, the estimate became smaller and more uncertain (OR = 1.300, 95% CI [.792, 2.135], $p = .29898$). The sample contained no misaligned corporate or partnered video, so relation and source were incompletely crossed. The smaller adjusted estimate shows that the initial relation–yield pattern was sensitive to the observed source composition; it does not establish mediation.

The base other/ambiguous-versus-aligned contrast was also positive (OR = 1.980, 95% CI [1.364, 2.874], $p = .00033$). This non-focal category is distinct from the focal misaligned-versus-aligned comparison.

Claim-visibility patterns were source-sensitive. Attributed or reputational claims accompanied lower substantive-comment yield than no clear claim in the base model (OR = .493, 95% CI [.290, .838], $p = .00897$, $q = .03899$), while the source-adjusted contrast was inconclusive (OR = .584, 95% CI [.358, .953], $p = .03136$, $q = .10977$). Table 3 reports the source-adjusted framing estimates discussed above.

Table 3*Source-Adjusted Framing Regression for Substantive-Comment Yield*

Framing indicator	B (standard error)	Odds ratio (95% confidence interval)	p	BH q	Conclusion
Diagnostic	.6459 (.2272)	1.908 [1.222, 2.978]	.00446	.02445	Higher substantive-comment yield; statistically significant after BH adjustment.
Prognostic	.0710 (.1883)	1.074 [.742, 1.553]	.70629	.81645	No clear difference.
Motivational	-.1684 (.1893)	.845 [.583, 1.225]	.37359	.57405	No clear difference.

Note. B is the log-odds coefficient; the value in parentheses is its standard error. p is the unadjusted coefficient-level p value. BH q is its Benjamini–Hochberg-adjusted value across the 126 fixed-effect coefficient tests in the 14 main regressions; $q < .05$ indicates statistical significance after that adjustment. The overall framing test p is a separate likelihood-ratio test. Source: Author’s analysis of the final analytical outputs.

All three framing indicators entered the same source-adjusted model. Each indicator compares videos with that frame to videos without the same frame while source form and the five shared controls are held constant. Taken together, the three indicators improved the model (overall framing test $p = .03837$).

The substantive-comment-yield results place greatest weight on diagnostic framing and source form. The relation estimate is source-sensitive under incomplete category overlap, and the associations may also reflect differences in issue and audience composition.

4.4 Conditional Firm Criticism and Context-Aware Negative Sentiment

Section 4.4 reports two separate comment outcomes. Conditional firm criticism is one dependent variable; represented relation, creator stance, represented conduct domain, and source form are tested as its independent variables. Context-aware negative sentiment is the second dependent variable; represented relation is its focal independent variable, with source form added in the adjusted model.

The planned primary comparison pointed toward more conditional firm criticism for misaligned than aligned videos. The estimated odds ratio was 1.594 using 1,893 firm-critical comments among 3,260 stable Tier-A firm evaluations across 186 videos and 38 firms. The wide interval (95% CI [.909, 2.795], $p = .10359$; overall represented-relation test $p = .18225$) leaves the size and certainty of the contrast unresolved, although the estimated direction is

upward. Appendix Table I1 reports restricted variants that qualify this estimate without replacing it.

Creator stance showed a clearer difference within the same eligible outcome. Critical or skeptical videos had higher modeled odds of conditional firm criticism than supportive videos (OR = 2.073, 95% CI [1.298, 3.309], $p = .00227$, $q = .01362$; overall creator-stance test $p = .00948$). The pooled counts were 744/1,082 and 567/1,216. Conditional firm criticism appeared in 55 of 67 eligible supportive threads. In a separate targeted set of favorable or mixed comments, 119 of the 457 classifiable comments contained an explicit caveat; this estimate applies only to that targeted set (Appendix Table H2).

The source-adjusted estimate for context-aware negative sentiment was higher for misaligned than aligned videos (OR = 1.782, 95% CI [1.114, 2.851], $p = .01603$; overall represented-relation test $p = .01696$), but it was not statistically significant after adjustment for multiple comparisons. The model used 4,105 negative labels among 8,898 valid Tier-A/B sentiment labels across 209 videos and 38 firms. This outcome covers negative expression toward any object and is distinct from conditional firm criticism.

For the secondary other/ambiguous-versus-aligned contrast, the source-adjusted context-aware negative-sentiment model (OR = 1.701, 95% CI [1.145, 2.526], $p = .00848$, $q = .03880$) and the negative-or-mixed reaction sensitivity (OR = 1.589, 95% CI [1.125, 2.244], $p = .00862$, $q = .03880$) remained statistically significant after adjustment. These secondary contrasts are distinct from the planned misaligned-versus-aligned comparison. Because the other/ambiguous category combines heterogeneous presentations and was not focal, the estimates are not treated as headline findings.

Table 4 summarizes these results. Figure 4 presents substantive-comment yield (Panel A), conditional firm criticism (Panel B), and context-aware negative sentiment (Panel C), each with its own eligible denominator and reference categories. Appendix Figure C1 shows adjusted probability profiles, and Appendix Table H5 reports the complete focal contrasts. All

rows report p values; additional contrasts also report Benjamini–Hochberg-adjusted q values across 126 fixed-effect tests. The F02 relation–yield sensitivity is evaluated with its coefficient p value.

Table 4
Selected Regression Results for Conditional Firm Criticism and Context-Aware Negative Sentiment

Outcome or model run	Video characteristic or comparison	Exact result	Overall characteristic test p	Conclusion
Panel A. Dependent variable: conditional firm criticism				
Conditional firm criticism	Misaligned rather than aligned	B = .4663; standard error = .2865; odds ratio = 1.594; 95% confidence interval [.909, 2.795]; p = .10359	p = .18225	No clear difference.
Conditional firm criticism	Critical/skeptical rather than supportive stance	B = .7288; standard error = .2388; odds ratio = 2.073; 95% confidence interval [1.298, 3.309]; p = .00227; BH q = .01362	p = .00948	Higher conditional firm criticism; statistically significant after BH adjustment.
Conditional firm criticism	Represented conduct domain	Communication/claim vs operational/environmental: OR = 1.859, p = .1042, BH q = .2371; product/service vs operational/environmental: OR = .809, p = .5138, BH q = .6961; strategy/governance/finance vs operational/environmental: OR = 1.402, p = .1460, BH q = .2968.	p = .15159	No clear overall difference.
Conditional firm criticism	Source form	Corporate/partnered vs independent commentary/platform material: OR = .780, p = .4099, BH q = .5955; news summary vs independent: OR = 1.460, p = .1786, BH q = .3462; personal testimony vs independent: OR = .667, p = .1422, BH q = .2936.	p = .11007	No clear overall difference.
Panel B. Dependent variable: context-aware negative sentiment				
Base relation model	Misaligned rather than aligned	B = .4690; standard error = .2264; odds ratio = 1.598; 95% confidence interval [1.026, 2.491]; p = .03831; BH q = .12704	—	Estimate above 1; not statistically significant after BH adjustment.
Relation model with source form held constant	Misaligned rather than aligned	B = .5776; standard error = .2398; odds ratio = 1.782; 95% confidence interval [1.114, 2.851]; p = .01603; BH q = .06171	p = .01696	Estimate above 1; not statistically significant after BH adjustment.
Outcome-definition check: negative or mixed reaction	Misaligned rather than aligned	B = .4676; standard error = .2200; odds ratio = 1.596; 95% confidence interval [1.037, 2.457]; p = .03356; BH q = .11428	—	Estimate above 1; not statistically significant after BH adjustment.

Note. Table 4 reports the primary misaligned-versus-aligned contrasts, creator-stance coefficient, and source-form and conduct-domain model summaries discussed in §4.4. Figure 4 shows M1–M3 estimates; Appendix Table H5 reports the full M2/M3 focal contrasts. B is the log-odds coefficient; SE is the standard error. p is unadjusted; BH q is the Benjamini–Hochberg-adjusted value for additional contrasts. Source: Author’s analysis of the final analytical outputs.

Figure 4

Video Characteristics Associated with Substantive-Comment Yield, Conditional Firm Criticism, and Context-Aware Negative Sentiment

Panel A. Substantive-comment yield

7,302 of 13,235 retrieved comments; 216 videos; 40 firms

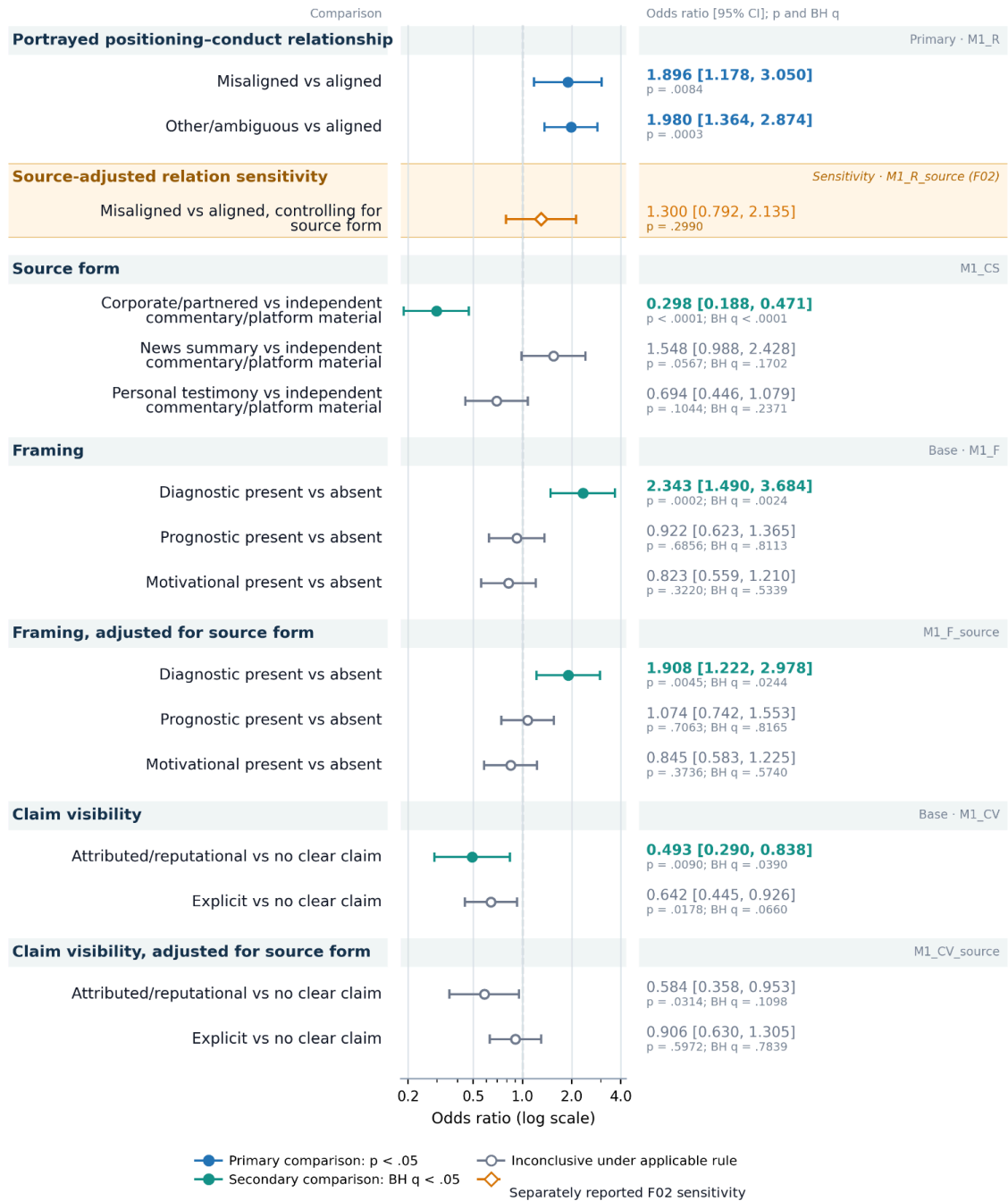
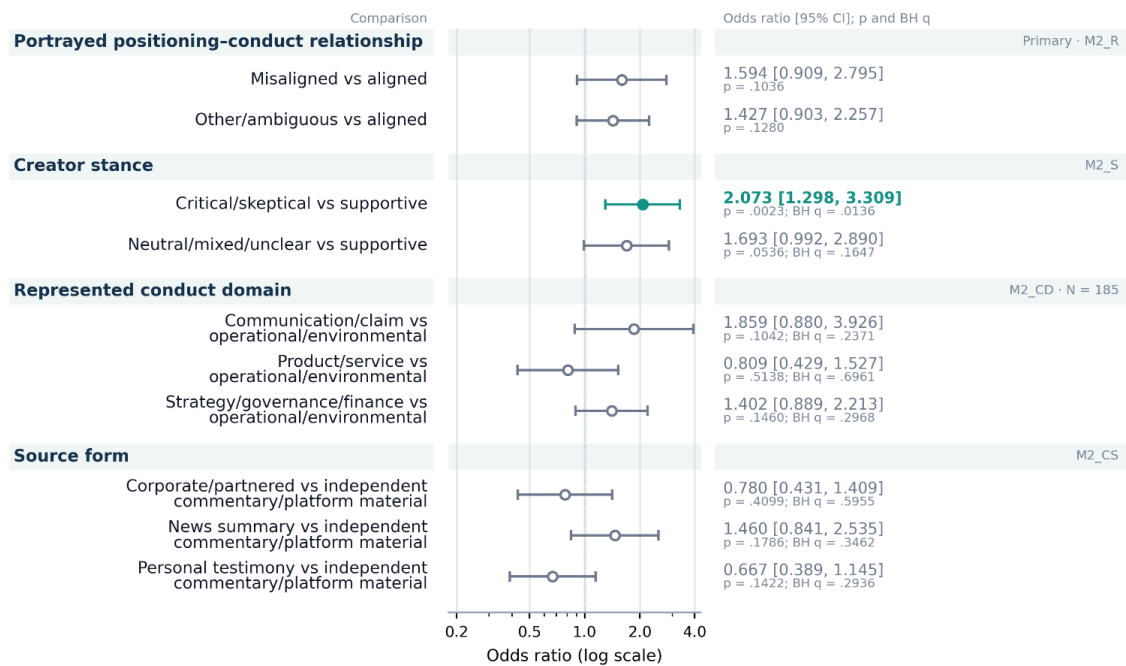


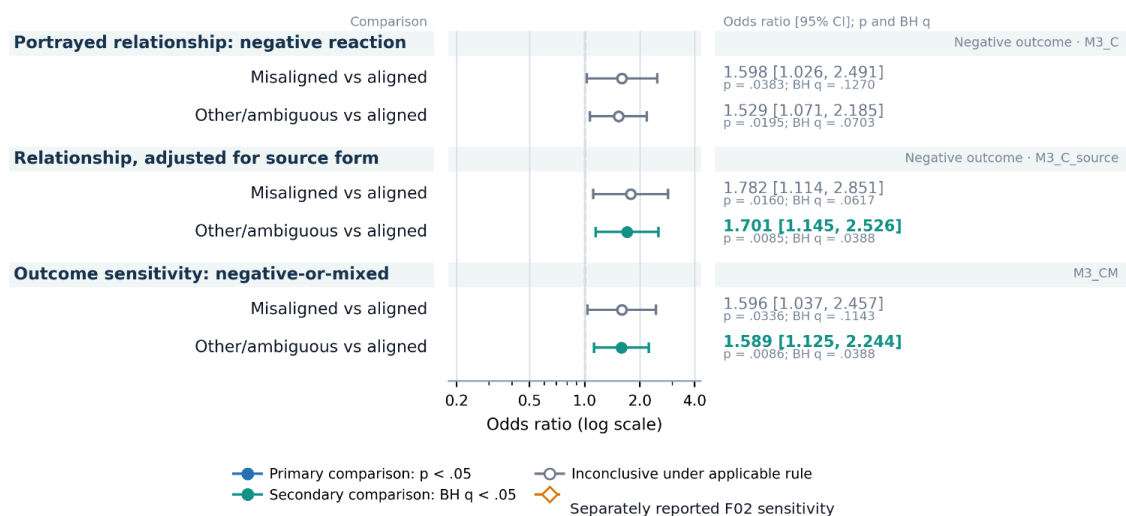
Figure 4 (continued)

Panel B. Conditional firm criticism

1,893 of 3,260 eligible comments; 186 videos; 38 firms (conduct-domain model: 1,893 of 3,256; 185 videos)

**Panel C. Context-aware negative reaction**

Negative: 4,105 of 8,898 eligible comments; negative-or-mixed sensitivity: 4,378 of 9,171; 209 videos; 38 firms



Note. Points show odds ratios with 95% Wald confidence intervals; the vertical line marks OR = 1. Across the thesis, the regression analysis comprises 14 main regression specifications plus one separately reported source-adjusted sensitivity (M1_R_source; F02). Panels A–C show the 13 M1–M3 regressions and F02; the exploratory non-firm primary-target-rate model M4_RA is reported in §4.5 and Table 5. BH q values reflect adjustment across the 126 fixed-effect tests in the 14 main regressions; F02 is shown with its coefficient p value. Filled colored points meet the applicable p or BH q threshold; hollow gray points are not statistically significant under that threshold. Each outcome retains its own eligible base. Appendix Table H5 provides the complete focal M2/M3 numerical results. Author's analysis of the final analytical outputs.

4.5 Credibility Challenges, Explicit Expectations, Action-Oriented Wording, and Primary Evaluative Target

Section 4.5 presents three descriptive profiles—credibility challenges, explicit expectations and action-oriented wording, and primary evaluative target—and one regression. The descriptive profiles do not have dependent- and independent-variable roles. The regression uses the non-firm primary-target rate as its dependent variable and represented relation as its independent variable.

The credibility-challenge rates below are descriptive rather than regression estimates. Claim visibility was not entered as an independent variable in a credibility-challenge regression, so no regression coefficient, standard error, or odds ratio applies. Credibility challenges appeared when commenters questioned, rejected, or conditionally contested an identifiable claim. Among 7,302 Tier-A comments, 557 did so across 93 threads and 30 firms. Challenge rates were 12.4% for explicit claims, 10.9% for attributed or reputational claims, and 1.4% when no clear claim was visible (426/3,448; 87/796; 44/3,058). These profiles remain descriptive because source, issue, and represented relation also differed across the groups.

Of the 557 challenges, 470 explicitly compared positioning with conduct shown or described. The pattern identifies a specific claim–conduct question for independent factual assessment. Most challenges diagnosed a concern and left the next step open. An explicit expectation, action-oriented wording, or both appeared in 93/557 challenges (16.7%); the remaining 464/557 (83.3%) stated neither an explicit expectation nor action-oriented wording (2,000-draw firm-bootstrap 95% CI [80.32%, 86.82%]). Figure 5 and Appendix H report the separate explicit-expectation, action-oriented-wording, and overlap counts and their broader eligible bases.

Evaluation also extended beyond the focal firm. Among 6,406 Tier-A comments with a stable primary evaluative target, 3,260 targeted the firm and 3,146 targeted an institution or system, consumers, or the creator. Another 890 other or unclear assignments and six unstable

assignments remained outside that base. In an exploratory model, misalignment accompanied higher odds of a non-firm target (OR = 2.189, 95% CI [1.283, 3.735], $p = .00406$, $q = .02328$; overall represented-relation test $p = .000275$). Table 5 places this exploratory regression beside the descriptive credibility-challenge rates while keeping the two panels distinct.

Table 5

Descriptive Credibility-Challenge Rates and the Exploratory Non-Firm Primary-Target-Rate Regression

Panel A. Descriptive credibility-challenge rates

Claim visible in the video	Credibility challenges / Tier-A comments	Descriptive challenge rate
Explicit claim	426 / 3,448	12.4%
Attributed or reputational claim	87 / 796	10.9%
No clear claim	44 / 3,058	1.4%

Panel B. Exploratory regression: dependent variable = non-firm primary-target rate; independent variable = represented relation

Represented-relation comparison	B (standard error)	Odds ratio (95% confidence interval)	p	BH q	Interpretation
Misaligned rather than aligned	.7833 (.2726)	2.189 [1.283, 3.735]	.00406	.02328	Higher non-firm targeting; exploratory.
Other/ambiguous rather than aligned	.8859 (.2222)	2.425 [1.569, 3.749]	.000067	.00106	Higher non-firm targeting; exploratory.

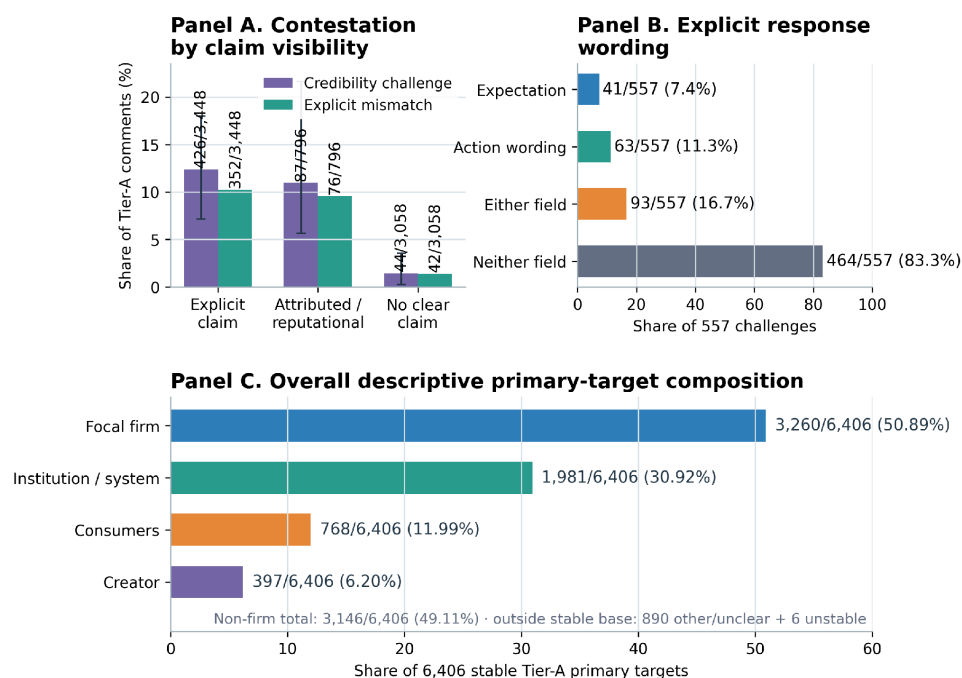
Note. Panel A of Table 5 reports descriptive rates, not a regression. Panel B of Table 5 reports the exploratory represented-relation regression; the two panels answer different questions. Source: Author's analysis of the final analytical dataset.

Figure 5 places three questions side by side: whether a claim was challenged, whether the challenge contained an explicit expectation or action-oriented wording, and which actor was the primary evaluative target. Each panel uses the comments able to answer its question.

Figure 5

Credibility Challenges, Explicit Expectations, Action-Oriented Wording, and Primary Evaluative Target

Credibility, response specification, and evaluative target



Note. Panel A of Figure 5 reports credibility challenges by claim visibility; Panel B of Figure 5 reports explicit expectations and action-oriented wording within 557 credibility challenges; Panel C of Figure 5 reports primary evaluative target among 6,406 stable Tier-A targets. Firm-bootstrap intervals use 2,000 draws. The panels use different eligible bases. Explicit mismatch is a comment-level code and is distinct from video-level represented-relation misalignment. Appendix H gives the full counts and adjusted profiles. Author's analysis of the final analytical dataset.

Together, these results show two features of commenter evaluation: credibility challenges usually expressed a diagnosis without an explicit next step, and scrutiny often extended beyond the firm. The exploratory misalignment–target association requires confirmation in a new sample.

4.6 Qualitative Contrasts, Measurement Check, and Model Checks

The 29 deliberately selected cases cover 29 videos, 27 firms, and 501 selected comments. The two contrasts below compare threads with similar coded presentation features but different criticism patterns and evaluated actors. They illustrate possible explanations and boundaries; they do not estimate prevalence.

Two electric-vehicle cases shared supportive, aligned, and independent-commentary codes but showed opposite firm-directed patterns. All eight eligible comments in the selected Uber thread opposed the firm; none of eight did so in Ford. Uber comments centered on costs borne by drivers; Ford comments centered on manufacturing cost, range, infrastructure, and competition. Because source, stance, and relation codes were the same, the contrast points to the concrete problem attached to the firm: driver burden in the Uber thread and product feasibility in the Ford thread.

A second pair shows what target coding adds. In the selected 3M thread, conditional firm criticism was 5/5 and non-firm targets 5/10; the corresponding Occidental counts were 4/16 and 32/48. The contrast separates scrutiny of company conduct from scrutiny distributed across the company, regulators, and the wider system.

A separate measurement check compared labels assigned by the published classifier from the written comment alone with researcher-reviewed labels that also used video and thread context. Among 222 eligible comments from a targeted set of 228, overall agreement was .6577 and Cohen's κ was .4738. Of the 128 comments labeled negative with context, the classifier labeled 73 negative, 52 neutral, and three positive. The largest observed difference therefore concerned comments labeled negative with video or thread context but neutral when only the written comment was supplied. Because the comparison also used different labeling procedures, it does not isolate a context effect or treat either label set as ground truth. Because the set was targeted, these figures describe this comparison and not the classifier's general performance.

The available category combinations also shaped model choice. No misaligned video was coded as corporate or partnered material, and each had a critical/skeptical or neutral/mixed/unclear creator stance; only two aligned videos were critical or skeptical, falling to one in the criticism base. Because some combinations were rare or absent, the study used separate models matched to the comparisons supported by the sample. Larger models did not

improve fit or prediction across firms. Appendix H reports the exact checks. Limited exploratory within-between checks in Appendix D were directionally consistent with the source and stance findings, while the relation–criticism contrast remained uncertain.

The final analysis contains 14 regressions: six for substantive-comment yield, four for conditional firm criticism, three for context-aware negative sentiment including the negative-or-mixed reaction outcome-definition check, and one for whether a non-firm actor was the primary evaluative target. Results for source-adjusted claim visibility and the source-form and represented-conduct-domain comparisons for conditional firm criticism were inconclusive. Appendix Table H4 lists all 14 main regressions and their roles. The source-adjusted represented-relation–yield check in §4.3 was an additional sensitivity analysis.

These checks determine how much weight each finding receives in the discussion. Diagnostic framing, source form, and creator stance carry the most weight. The evidence for misalignment remains inconclusive. Claim and target coding show what was disputed and whom or what was evaluated; the misalignment–target comparison remains exploratory.

5. Discussion and Conclusion

Across the selected-video records, creator presentations of corporate sustainability varied by source form. Creators most often presented the issue through independent commentary or platform material, and critical or skeptical stances were slightly more common than supportive ones. Most videos defined a sustainability problem, about two-thirds displayed an explicit or attributed claim, and misalignment was less common than alignment or ambiguity.

In the 216 linked threads, diagnostic framing accompanied higher substantive-comment yield, corporate or partnered source form accompanied lower substantive-comment yield than independent commentary or platform material, and critical or skeptical creator stance accompanied higher conditional firm criticism than supportive stance. Platform engagement rate showed no clear monotonic association with substantive-comment

yield. Misalignment remained inconclusive: its substantive-comment-yield association was sensitive to source adjustment; the planned conditional-firm-criticism comparison remained uncertain; the source-adjusted comparison for context-aware negative sentiment was not statistically significant after adjustment for multiple comparisons; and the represented-relation comparison for non-firm primary-target rate remained exploratory.

The comments added two further distinctions. Of 557 credibility challenges, 470 compared a sustainability proposition with portrayed conduct, while 464 stated neither an explicit expectation nor action-oriented wording. Among 6,406 comments with a stable primary evaluative target, 3,146 (49.1%) evaluated an actor other than the focal firm. Together, the findings answer the research question at two levels: how creators presented the firm and issue, and what commenters discussed, criticized, or challenged beneath those presentations. These observational results apply to the selected threads.

5.1 What the Findings Mean

The three supported associations concern different response layers: source form and diagnostic framing accompanied substantive-comment yield, whereas creator stance accompanied conditional firm criticism. The near-zero platform-engagement–yield association further distinguishes visible activity from comment content, complementing evidence that different interaction counts can reflect different message features (Cheng & Li, 2024; Jha & Verma, 2024).

Diagnostic framing was associated with higher substantive-comment yield. Benford and Snow (2000) define diagnostic framing as identifying a problem, cause, or implicated actor, while Edgerly et al. (2013) found that topics mentioned in YouTube videos tended to reappear in their comments. Together, this literature suggests one possible explanation: diagnostic framing may give commenters a clearer issue to address. The observed association supplies the pattern; a controlled design can test the explanation.

Corporate or partnered source form accompanied lower substantive-comment yield than independent commentary or platform material. Gómez-Carrasco et al. (2021) found that firm insiders emphasized more supplementary and favorable CSR topics, whereas outside stakeholders focused more on core CSR issues. Dunn and Harness (2019) showed that source cues shape how company- and user-generated CSR communication is received. This literature suggests two possible explanations for the present association: differences in issue emphasis and differences in how viewers interpret the source. Audience composition and exposure provide further explanations to test. Diagnostic framing remained positively associated in a separate source-adjusted model, supporting the treatment of source form and diagnostic framing as distinct video characteristics.

Critical or skeptical creator stance accompanied more conditional firm criticism than supportive stance. One possible explanation is that a critical presentation foregrounds the firm as an object of evaluation. The Uber/Ford contrast described in Section 4.6 and documented in Appendix Table E1 illustrates why stance alone remains incomplete: two supportive and aligned videos produced different criticism patterns while commenters focused on different practical concerns. Creator stance identifies the direction of presentation; the issue raised in the thread identifies what the criticism concerns.

The misalignment results did not support a clear conclusion. The planned conditional firm criticism estimate was imprecise, and its confidence interval crossed 1; the source-adjusted misaligned-versus-aligned context-aware negative sentiment comparison did not meet the multiple-testing-adjusted threshold. No corporate or partnered video was coded as misaligned, leaving source form and represented relation incompletely separated. These limits require a cautious interpretation of the observed direction.

Among the coded comments, 557 challenged an identifiable proposition, 470 compared a proposition with portrayed conduct, and 464 stated neither an explicit expectation nor action-oriented wording. Lyon and Montgomery (2013) theorize that social media exposes

corporate environmental communication to public challenge, while Liou et al. (2026) found that CSR-related online discussion can reveal perceived gaps between corporate conduct and public expectations. The present evidence adds a further distinction: recognizing a possible proposition–conduct gap and expressing an expectation or proposed action are separate forms of comment expression. Future research can test what follows these challenges. This study also found that institutions, systems, consumers, and creators together received almost half of the stable primary evaluative targets. The primary evaluative target identifies whom or what was evaluated; the firm’s connection, relevant evidence, responsibility, influence, authority to act, and any expected action remain separate questions.

5.2 Research Contributions

The main academic contribution is a layered account of video–comment response. Edgerly et al. (2013) found that video topics can reappear in comments. This thesis extends that relationship by showing that different video features corresponded with different response layers: source form and diagnostic framing were associated with substantive-comment yield, whereas creator stance was associated with conditional firm criticism. Platform engagement rate did not clearly indicate whether comments focused on sustainability in this corpus. Liou et al. (2026) found that CSR-related online discussion can reveal perceived conduct–expectation gaps. This thesis further separates credibility challenge, proposition–conduct comparison, explicit expectation, and action-oriented wording: 464 of 557 credibility challenges contained neither an explicit expectation nor action-oriented wording. Whether challenges are followed by expectations or requested actions remains a question for future research.

Methodologically, the study keeps each comment linked to its video context and uses a separate eligible base for each outcome. Firm grouping, category-support checks, field-specific recoding, the targeted comparison with a published sentiment classifier, and qualitative cases answer different validation questions. Two reproducibility checks addressed

different objects: the classifier rerun reproduced all 13,235 classifier labels and the reported statistics for the 222-comment comparison, while predictor-blinded recoding measured run-to-run consistency for selected AI-coded fields. This design makes each conclusion and its evidence base traceable.

For organizations, the same linked evidence supports the two practical uses developed next: assessing communication against its intended outcome and defining the concern before deciding what requires review.

5.3 Organizational Implications

Firms can define the sustainability position they intend to communicate, but that does not determine how their positioning and portrayed conduct are evaluated in public discourse. Linked video–comment analysis offers a structured way to examine the part of that discourse visible in selected social-media threads. It shows which issues are discussed, where comments raise a possible gap between a sustainability proposition and portrayed conduct, whom or what they evaluate, and whether they state an explicit expectation or use action-oriented wording. With sampling, coding, and validation designed for the purpose, the approach could be applied to one firm, selected competitors, an industry, another corporate issue, or another platform. The analytical structure is adaptable; the present findings remain specific to selected sustainability-related TikTok threads.

For communication decisions, firms can first define the intended outcome. Platform engagement rate showed no clear monotonic association with substantive-comment yield. Video views and platform activity may matter when visibility or interaction is the goal, but they do not show whether linked comments addressed the sustainability issue. Corporate or partnered source form accompanied lower yield than independent commentary or platform material, whereas diagnostic framing accompanied higher yield. Firms can therefore evaluate communication and partnerships against the intended outcome. When issue-focused discussion is intended, they can test content that clearly explains the problem, its cause, or

responsibility and examine whether linked comments address that issue. These observational associations identify features to compare and test, not proven causes.

Comment analysis can then define the concern. Critical or skeptical stance accompanied higher conditional firm criticism than supportive stance, showing why criticism should be interpreted with the video that prompted it. Of 557 credibility challenges, 470 compared a sustainability proposition with portrayed conduct. Examining the proposition and portrayed conduct shows what commenters treated as inconsistent and what may require factual review; it does not establish that the concern is accurate. In addition, 3,146 of 6,406 stable primary targets concerned a non-firm actor or system. A thread may therefore center on the firm, another actor, or a wider system. The target shows where the discussion places the issue; the firm's role depends on evidence about its connection, responsibility, influence, and capacity to act. The Uber/Ford contrast in §4.6 illustrates why similar high-level codes can point to different practical concerns.

Among the same 557 credibility challenges, 464 contained neither an explicit expectation nor action-oriented wording. Comments often identified the concern, not the remedy. Verified evidence, materiality, business relevance, and strategy can then guide whether the firm clarifies, investigates, adapts, engages, coordinates, monitors, or takes no further action. Table 6 summarizes this proposed use.

Table 6
How Linked Video–Comment Analysis Can Support Organizational Decisions

Strategic question	Evidence to examine	Why it matters
1. How is the firm positioned and its conduct evaluated?	Issue discussed; sustainability proposition; portrayed conduct; primary evaluative target; any explicit expectation or action-oriented wording.	Defines a perceived positioning–conduct concern in the selected discourse and shows whether evaluation centers on the firm, another actor, or a wider system.
2. Did communication serve its intended goal?	Communication objective; video views; platform engagement rate; substantive-comment yield; source form; diagnostic framing.	Assesses communication and partnerships against the intended outcome and supports testing clearer issue explanations when issue-focused discussion is intended.
3. What requires review?	Specific concern; verified evidence; materiality; business relevance; firm connection, responsibility, influence, and capacity to act.	Defines a focused evidence question and supports a reasoned decision on whether and how to respond.

Note. These are proposed uses requiring purpose-designed sampling, organizational judgment, and prospective testing. The present findings concern selected sustainability-related TikTok

threads; applying the approach to another firm set, industry, issue, or platform requires new sampling and validation. The primary evaluative target identifies whom or what was evaluated, not who is responsible or able to act. Appendix Table K1 states the related evidence requirements and interpretation boundaries. Source: Author's organizational synthesis based on the study findings.

5.4 Limitations

This study is bounded to public, search-visible English-language TikTok videos involving large U.S. firms and the direct comments available at collection. Search terms, platform ranking, screening, the 200-comment cap, exclusion of replies, and available metadata shaped the observed material. Researcher review covered all 314 shortlisted candidates, but it did not estimate whether prioritization omitted eligible videos elsewhere in the 5,210-URL pool. Video-side descriptions cover all 217 selected records; linked comment analyses characterize the 216 usable threads and the expressions visible within them.

Source form, creator stance, framing, represented conduct domain, and represented relation often co-occurred, and some combinations were rare or absent. The corpus supports associations, not mechanisms; balanced or experimental designs are needed to test them.

Coding reduces complex multimodal material to categories. The primary-target variable simplifies some multi-actor comments, and the 29 qualitative cases illustrate patterns rather than prevalence. Predictor-blinded recoding measured one configured AI system's run-to-run consistency, not accuracy or agreement with independent human coders; the targeted classifier comparison did not validate other fields. The strongest conclusions therefore concern what was presented, discussed, challenged, and evaluated in selected threads. External organizational and operational records are needed to establish underlying facts and material importance.

Future replications should include independent human double-coding of a probability sample for the main comment fields. That check would complement, rather than replace, the present run-to-run reproducibility evidence and targeted sentiment-classifier comparison.

5.5 Future Research

Preregistered experiments could hold issue, evidence, length, and production quality constant while varying source cue, diagnostic framing, or creator stance. Outcomes could follow the response sequence—willingness to comment, substantive-comment yield, and conditional firm criticism—while issue clarity, source independence, credibility, and evidence strength test possible mechanisms.

Prospective thread research could test whether challenges are followed by expectations or requested actions. Reply chains collected at fixed intervals could record the proposition–conduct comparison, expected responder, requested action, later clarification or dispute, and distinctions among the evaluated actor, responsible actor, evidence holder, and actor able to act. Building on Liou et al. (2026), this would extend the diagnostic–prognostic distinction to commenter discourse.

5.6 Conclusion

This thesis linked 216 sustainability-related TikTok video–comment threads involving 40 S&P 500 firms and 13,235 direct comments. Three associations were clear: diagnostic framing with higher substantive-comment yield; corporate or partnered source form with lower yield than independent commentary or platform material; and critical or skeptical stance with higher conditional firm criticism than supportive stance. Platform engagement rate showed no clear association with yield. Credibility challenges often compared propositions with portrayed conduct; expectations and action-oriented wording were uncommon; and primary targets often lay beyond the focal firm. These findings describe selected public English-language threads—not wider public opinion or verified firm conduct. The linked design helps researchers distinguish activity from evaluation; for organizations, it supports examining firm positioning in selected discourse, assessing communication against its goal, and defining what requires review. Public discourse can inform review; verified evidence, materiality, and the firm’s role determine what follows.

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Appendices

These appendices retain the validation detail, denominator records, robustness checks, case boundaries, supplementary profiles, and disclosure needed to interpret the main analysis. Together, they extend the descriptive and methodological detail supporting the main findings while preserving the thesis's evidentiary distinctions.

Appendix A. Platform Engagement Rate and Substantive-Comment Yield

Figure 3 reports $\rho = .0376$ ($p = .583$; 2,000-draw firm-bootstrap 95% CI $[-.147, .245]$) across 216 comment-bearing videos. The near-zero estimate remains compatible with modest monotonic associations in either direction and does not establish a platform-wide relationship. Appendix Table A1 completes the four-quadrant video-count cross-check and reports the two opposite-quadrant comment aggregates.

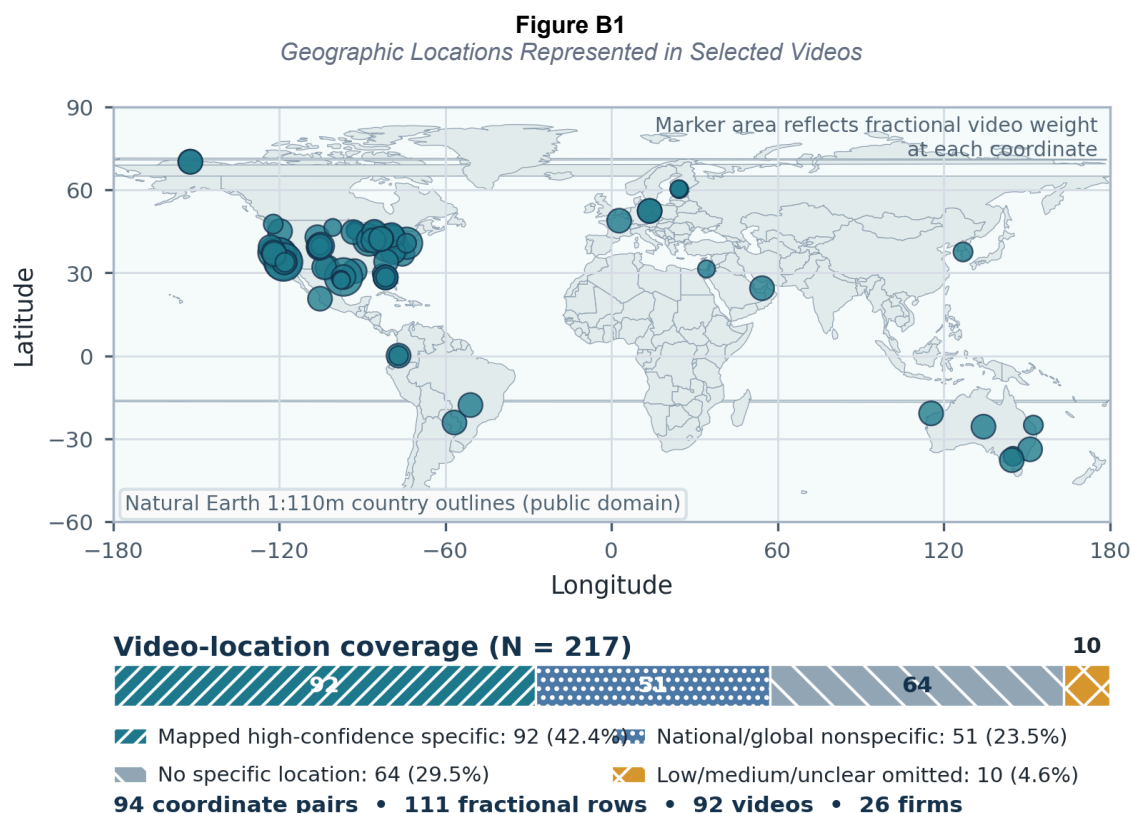
Table A1
Descriptive Platform-Engagement–Yield Quadrants

Quadrant	Observed count / comment aggregate	Use
Above-median platform engagement rate / above-median substantive-comment yield	56 videos; pooled comment aggregate not reported	Cell shown in Figure 3
Above-median platform engagement rate / below-median substantive-comment yield	52 videos; 1,694 Tier-A comments / 4,451 retrieved comments (38.1%)	Opposite quadrant discussed in §4.2
Below-median platform engagement rate / above-median substantive-comment yield	52 videos from 25 firms; 2,042 Tier-A comments / 2,598 retrieved comments (78.6%)	Opposite quadrant discussed in §4.2
Below-median platform engagement rate / below-median substantive-comment yield	56 videos; pooled comment aggregate not reported	Cell shown in Figure 3

Note. The four video counts total 216. Tier A denotes substantive comments under §3.3. Only the two opposite quadrants discussed in §4.2 receive pooled comment aggregates here; no additional pooled aggregate is introduced for the other two cells. “Above median” and “below median” denote sample partitions, not natural, quality, managerial, or early-warning thresholds. Author’s analysis of the final analytical dataset.

Appendix B. Locations Represented in Selected Videos

Figure B1 shows the geographic breadth represented in the full selected-video record and provides context for the later quantitative findings. High-confidence coordinates were identified for 92 of 217 selected-video records (42.4%) and for 92 of the 150 videos separately classified as representing a specific location (61.3%), spanning 26 firms. One of the 217 records did not yield a usable direct-comment thread; the linked video–comment analyses therefore use 216 threads. The map contains 111 fractionally weighted location rows at 94 unique coordinate pairs, with weights summing to 92 videos. It shows where selected videos situated their content, not commenter residence, geographic prevalence, or the full geography of firm operations.



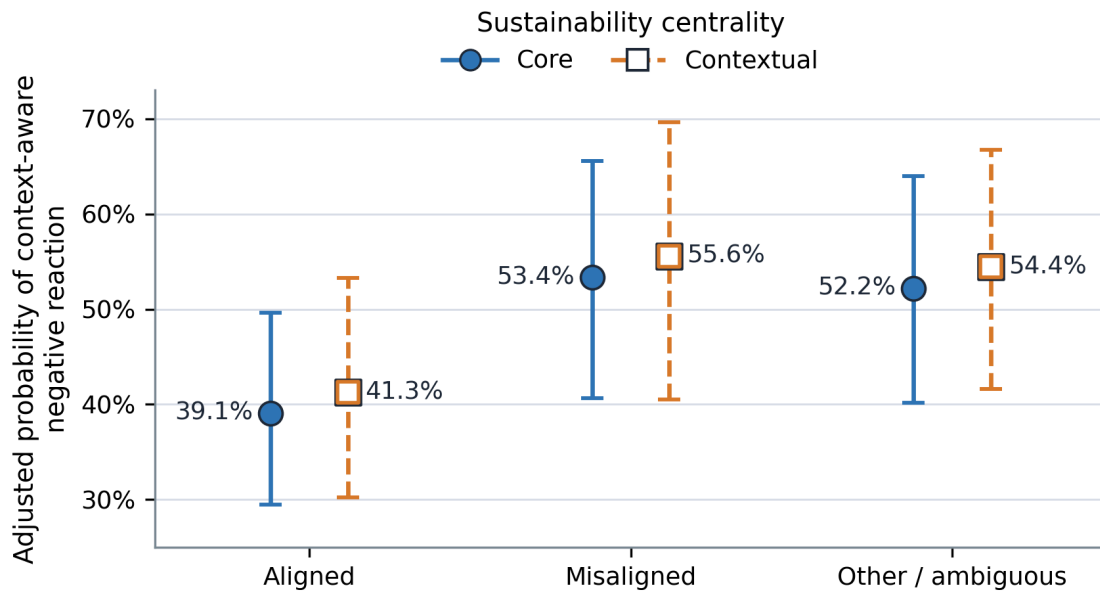
Note. The map represents 111 fractionally weighted location rows at 94 unique coordinate pairs from 92 selected videos and 26 firms. Symbol size reflects fractional video weight; coverage bars use the full-video denominator of 217. Country outlines use world-atlas 2.0.2, a TopoJSON redistribution of Natural Earth vector data version 4.1.0 (Bostock, 2019); points use direct longitude–latitude placement. Source: Author’s analysis.

Of 16 exploratory geography fits, 14 were valid; none was statistically significant after adjustment for multiple comparisons, and two invalid fits were excluded from inference. These fits form a separate exploratory family and are not part of the 14-regression main-analysis set described in §3.5 and Appendix Table H4. Figure B1 documents where the selected content was situated and identifies a starting point for a purpose-built geographic study.

Appendix C. Context-Aware Negative Sentiment

Context-aware negative sentiment comprises 4,105 negative classifications among 8,898 valid three-class context-linked classifications across 209 videos and 38 firms. The plotted probabilities come from the source-adjusted beta-binomial model, with aligned representation as the reference. This outcome captures affect toward any object, not specifically conditional firm criticism.

Figure C1
Context-Aware Negative Sentiment



Note. Points are adjusted probabilities with 95% Wald CIs for aligned, misaligned, and other/ambiguous represented relations at core and contextual sustainability centrality. Category labels follow the preserved model-output order (aligned, misaligned, other/ambiguous). In the source-adjusted model, misaligned versus aligned representation had OR = 1.782, 95% CI [1.114, 2.851], coefficient $p = .01603$, whole-characteristic likelihood-ratio $p = .01696$, and Benjamini–Hochberg-adjusted $q = .06171$. This q value comes from the 126 fixed-effect coefficient tests across the 14-regression main-analysis set. The positive point estimate was not statistically significant after that adjustment and therefore does not support a firm conclusion. It does not establish blame, trust loss, approval, reputational harm, population sentiment, or conditional firm criticism. Author's analysis of the final analytical dataset.

Appendix D. Observational Firm-Level Sensitivity Checks

Table D1 reproduces four archived observational firm-level sensitivity estimates and reports the limited number of firms represented in both categories of each comparison. The generating formula and script were not available in the current source package, so the estimates could not be independently reproduced. They are retained as appendix-only sensitivity evidence, not as causal fixed-effects estimates or replacements for the outcome-specific models. These four checks sit outside the 14-regression main-analysis set and do not create another outcome family.

Table D1
Observational Firm-Level Sensitivity Estimates

Outcome contrast	OR	95% CI	BH q	Strict pairwise firms, n
Misaligned versus aligned substantive-comment yield	1.844	[1.092, 3.113]	.0496	13
Corporate/partnered versus independent commentary/platform material substantive-comment yield	.363	[.204, .644]	.0049	10
Misaligned versus aligned conditional firm criticism	1.727	[.900, 3.313]	.1503	11
Critical/skeptical versus supportive creator stance, conditional firm criticism	2.049	[1.169, 3.592]	.0496	14

Note. OR = odds ratio; CI = confidence interval. “Strict pairwise firms” is the number of firms represented in both categories of the named contrast within the relevant outcome-specific sample. The BH q column reproduces the Benjamini–Hochberg-adjusted values in the archived sensitivity record; these appendix-only values are separate from the 126-test main-analysis adjustment. The source-form and creator-stance directions are consistent with the corresponding main results, while the represented-relation contrast for conditional firm criticism remains uncertain. Author’s analysis.

Appendix E. Qualitative Case Validation and Complete Case Registry

The qualitative layer contains 29 purposively selected video-thread/context units from 27 firms and 501 selected comments. Each unit links one video, its direct-comment thread, the full-thread counts used as quantitative denominators, a selected-comment interpretation, and an explicit inference boundary. Selection supports explanation and boundary identification, not prevalence, matching, or causal inference.

All 29 case interpretations received a separate review: 27 were confirmed, one was inconclusive, and one was corrected. Six cases also received additional human confirmation. These checks support the stated case interpretations and remain separate from the quantitative outcome coding. Table E1 presents selected case roles and boundaries; Appendix Table E2 retains the complete registry.

Table E1
Selected Qualitative Case Roles and Boundaries

Case unit(s)	Full-thread denominator chain; selected comments	Illustrative role	Boundary; confidence
Q27/V194 — Amazon; video coded as partnered; video-stated 88%-less-plastic refill claim	166 retrieved / 0 Tier A / 0 firm-directed; 18 selected	Visible platform interaction without issue-focused direct comments	One thread; no partnership/source effect or prevalence; high
Q08/V084 — Uber; Q10/V170 — Ford	Uber: 14 retrieved / 13 Tier A / 8 firm-directed; 8/8 against firm; 14 selected. Ford: 23 retrieved / 21 Tier A / 8 firm-directed; 0/8 against firm; 18 selected.	EV economic counter-framing and stakeholder-burden contrast	Paired, not matched or causal; Q08 core, Q10 contextual; both medium
Q14/V089 — 3M; Q18/V024 — Occidental	3M: 15 retrieved / 13 Tier A / 5 firm-directed / 10 stable targets; 5/5 against firm; 5/10 non-firm; 15 selected. Occidental: 59 retrieved / 48 Tier A / 16 firm-directed / 48 stable targets; 4/16 against firm; 32/48 non-firm; 18 selected.	Focal-firm versus non-firm evaluative routing	Structured, not formally matched; no adjudication, wrongdoing, responsibility, or prevalence; both high

Note. Full-thread counts provide the quantitative denominators. Selected comments and available textual video context support the qualitative interpretations; the review addressed case interpretations rather than a new full-thread recode or video rewatch. Reader-facing examples are privacy-screened paraphrases except for the three short direct comments in Figure 1, which retain archived wording after usernames and profile images were removed. Other exact text and identifiers remain restricted. Author's qualitative synthesis and confirmation records.

Table E2 records the selection role, interpretive mechanism, boundary, and review confidence for all 29 cases. It is a compact audit registry; the retained project case map preserves the authorized paraphrases, evidence bases, organizational-triage uses, and source lineage.

Table E2
Complete Qualitative Case Registry, Panel A (Q01–Q06)

Case / firm	Selection role / mechanism	Boundary; confidence and review
Q01 — NORFOLK SOUTHERN CORP	critical/skeptical creator stance / high conditional firm criticism; issue severity	Political and governmental blame means the case is not evidence that all accountability is focal-firm-only; allegations remain represented claims. Confidence: high; review: separately reviewed in the project record, with additional human confirmation.
Q02 — CONOCOPHILLIPS	critical/skeptical creator stance / high conditional firm criticism; creator cueing	Mobilization and repeated slogans do not independently establish project impacts or population-wide opposition. Confidence: high; review: separately reviewed in the project record.
Q03 — P G + E CORP	critical/skeptical creator stance / high conditional firm criticism; source-form pattern	Personal experiences are relevant signals but are not audited billing records; some comments redirect toward consumer choices or the green-energy industry. Confidence: high; review: separately reviewed in the project record.
Q04 — APPLE INC	critical/skeptical creator stance / low conditional firm criticism; responsibility diffusion	The case includes genuine partial defenses of Apple; final use must not collapse qualification into uniform criticism. Several comment-code doubts were documented but not selectively applied. Confidence: medium; review: separately reviewed in the project record, with additional human confirmation.
Q05 — PHILLIPS 66	critical/skeptical creator stance / low conditional firm criticism; issue severity	A closure-is-good interpretation is explicitly contested by a jobs-and-prices counterexample; local testimony does not itself verify causation. Confidence: medium; review: separately reviewed in the project record.
Q06 — TESLA INC	critical/skeptical creator stance / low conditional firm criticism; counter-framing	A smaller subset still criticizes Tesla environmental claims; the case does not establish that boycott concerns are factually correct or broadly shared. Confidence: medium; review: separately reviewed in the project record.

Table E2 (continued)
Complete Qualitative Case Registry, Panel B (Q07–Q12)

Case / firm	Selection role / mechanism	Boundary; confidence and review
Q07 — CUMMINS INC	supportive creator stance / high conditional firm criticism; counter-framing	A minority supports lower emissions if performance is preserved; criticism is not proof that the technology performs poorly. Confidence: high; review: separately reviewed in the project record.
Q08 — UBER TECHNOLOGIES INC	supportive creator stance / high conditional firm criticism; counter-framing	Final confidence is medium because the solution actor is inferred; no reviewed comment clearly endorses the program, but the case does not prove program harm. Confidence: medium; review: separately reviewed in the project record, with additional human confirmation.
Q09 — DELTA AIR LINES INC	supportive creator stance / high conditional firm criticism; prior reputation	Some selected material is meme-like or unrelated customer grievance; represented corrective action does not imply acceptance or causally produce skepticism. Confidence: high; review: separately reviewed in the project record.
Q10 — FORD MOTOR CO	supportive creator stance / low conditional firm criticism; responsibility diffusion	Comments run both ways on Ford's competitive prospects; the case is not an industry forecast. Confidence: medium; review: separately reviewed in the project record.
Q11 — WALT DISNEY CO/THE	supportive creator stance / low conditional firm criticism; source-form pattern	Substantial positive reaction remains; practical objections do not establish that the sustainability initiative fails overall. Confidence: medium; review: separately reviewed in the project record.
Q12 — STARBUCKS CORP	supportive creator stance / low conditional firm criticism; prior reputation	Supportive clarifications are prominent enough that the thread cannot be characterized as uniformly skeptical; skepticism does not verify material harm. Confidence: medium; review: separately reviewed in the project record.

Table E2 (continued)
Complete Qualitative Case Registry, Panel C (Q13–Q18)

Case / firm	Selection role / mechanism	Boundary; confidence and review
Q13 — ALPHABET INC CL A	neutral/mixed/unclear creator stance / high conditional firm criticism; counter-framing	Minority comments are neutral or techno-optimistic; demands do not establish actual local harm or the correctness of proposed regulation. Confidence: high; review: separately reviewed in the project record, with additional human confirmation.
Q14 — 3M CO	neutral/mixed/unclear creator stance / high conditional firm criticism; issue severity	Some comments redirect to other PFAS sources, taxes, or data centers; allegations and personal knowledge remain unverified represented content. Confidence: high; review: separately reviewed in the project record.
Q15 — BOEING CO/THE	neutral/mixed/unclear creator stance / high conditional firm criticism; counter-framing	The reviewed thread is small and includes technical clarifications and jokes; it cannot support broad audience conclusions. Confidence: medium; review: separately reviewed in the project record.
Q16 — MCDONALD S CORP	neutral/mixed/unclear creator stance / low conditional firm criticism; source-form pattern	Final solution responsibility is shared and confidence is medium; skepticism about theft or laziness does not reject reuse itself. Confidence: medium; review: separately reviewed in the project record, with additional human confirmation.
Q17 — XCEL ENERGY INC	neutral/mixed/unclear creator stance / low conditional firm criticism; issue severity	A strong safety defense is present; household losses do not by themselves show the shutoff was unnecessary, and safety support does not erase burden. Confidence: high; review: separately reviewed in the project record.
Q18 — OCCIDENTAL PETROLEUM CORP	neutral/mixed/unclear creator stance / low conditional firm criticism; responsibility diffusion	The case does not establish regulator or firm wrongdoing; some comments remain neutral or reputational rather than remedial. Confidence: high; review: separately reviewed in the project record.

Table E2 (continued)
Complete Qualitative Case Registry, Panel D (Q19–Q24)

Case / firm	Selection role / mechanism	Boundary; confidence and review
Q19 — XCEL ENERGY INC	high context-aware negative sentiment / low conditional firm criticism; issue severity	Negative sentiment is not identical to direct firm blame; explanatory comments do not necessarily defend the increase. Confidence: high; review: separately reviewed in the project record.
Q20 — VISTRA CORP	high context-aware negative sentiment / low conditional firm criticism; counter-framing	Some comments support a broader smart-grid portfolio; technology skepticism is not evidence that the specific project is unsafe. Confidence: high; review: separately reviewed in the project record.
Q21 — CHEVRON CORP	low context-aware negative sentiment / high conditional firm criticism; creator cueing	Many highly liked comments praise the creator or activist rather than directly evaluating Chevron; the case does not adjudicate the underlying litigation. Confidence: high; review: separately reviewed in the project record.
Q22 — DTE ENERGY COMPANY	low context-aware negative sentiment / high conditional firm criticism; prior reputation	Clear pro-governor and pro-clean-energy comments remain; redirection does not disprove the announced agreement. Confidence: medium; review: separately reviewed in the project record.
Q23 — UNITED AIRLINES HOLDINGS INC	low context-aware negative sentiment / high conditional firm criticism; counter-framing	A few comments are positive or information-seeking; pooled target allocation does not establish industry culpability or claim falsity. Confidence: high; review: separately reviewed in the project record.
Q24 — NVIDIA CORP	high non-firm primary-target rate; responsibility diffusion	The full-thread case-selection metric places stable targets outside the firm, but this is not quantitative validation of responsibility structure; technical defenses and support also appear. Confidence: high; review: separately reviewed in the project record, with additional human confirmation.

Table E2 (continued)
Complete Qualitative Case Registry, Panel E (Q25–Q29)

Case / firm	Selection role / mechanism	Boundary; confidence and review
Q25 — TJX COMPANIES INC	high non-firm primary-target rate; responsibility diffusion	Comments also report thrift-store overload, exclusion of poorer shoppers, and residual waste; the case does not establish that secondhand consumption is uniformly beneficial. Confidence: high; review: separately reviewed in the project record.
Q26 — CONOCOPHILLIPS	high non-firm primary-target rate; responsibility diffusion	Campaign participation is self-reported and the selected case does not establish prevalence, regulatory merit, or actual project impact. Confidence: high; review: separately reviewed in the project record.
Q27 — AMAZON.COM INC	corporate/partnered source form with low substantive-comment yield; limited issue-focused direct-comment discussion	Product or creator enthusiasm is not sustainability support; the thread supplies almost no responsibility allocation. Confidence: high; review: separately reviewed in the project record.
Q28 — MICROSOFT CORP	corporate/partnered source form with low substantive-comment yield; limited issue-focused direct-comment discussion	Unrelated product complaints are not sustainability criticism, and generic praise is not confirmation of environmental performance. Confidence: high; review: separately reviewed in the project record.
Q29 — DELL TECHNOLOGIES C	corporate/partnered source form with low substantive-comment yield; limited issue-focused direct-comment discussion	Bare purchase interest or rejection does not evaluate packaging, waste, or sustainability performance. Confidence: high; review: separately reviewed in the project record.

Note. The registry covers all 29 reviewed qualitative cases. Selection families and mechanisms organize interpretation; they do not estimate prevalence, causal effects, factual truth, or population response. The detailed Table E1 cases retain their full-thread denominator chains. Source: retained qualitative case map and final case-review record.

References to product claims, costs, burdens, regulation, or technology report what the sampled platform materials presented; this study did not verify their underlying factual truth.

Appendix F. Generative AI Assistance, Researcher Authority, Data Access, and Ethics

Ethical handling followed a context-sensitive approach rather than treating public accessibility as unrestricted permission to reuse material. The study did not contact users and limited analysis to the research questions. Reader-facing examples were generally paraphrased after privacy review; Figure 1 retains three short archived direct comments with usernames and profile images removed. Analytical files removed or minimized usernames, user IDs, profile links, and avatar information; exact text, captions, URLs, creator handles, stable identifiers, and linkage fields remained restricted when needed for verification (Franzke et al., 2020).

Generative AI and computational tools assisted with defined tasks, but they did not determine the research design or final conclusions. The researcher formulated the research questions, set the scope, developed and approved the codebook and eligibility rules, reviewed all 314 shortlisted videos, made every final inclusion decision, specified and checked the analyses, and approved the submitted interpretations and manuscript. This review selected 217 videos; 216 had usable direct-comment threads and formed the linked analytical corpus. AI-generated outputs were treated as proposals or structured coding inputs that could be revised, rejected, or excluded; the 13,235 comments were not manually adjudicated one by one.

Checks were matched to the task they could inform. Run-to-run consistency was checked through two separate predictor-blinded recoding runs by the same configured AI system on the same 360 comments; a published sentiment classifier served as an external benchmark for 222 eligible comments from a targeted 228-comment set; all 29 qualitative case interpretations received a separate review, with six receiving additional human confirmation; and statistical, source, numerical, and document checks addressed their named outputs. Table F1 links each task to the researcher decision and check that directly informed it. These task-specific checks do not mean that every one of the 13,235 comments was manually reviewed.

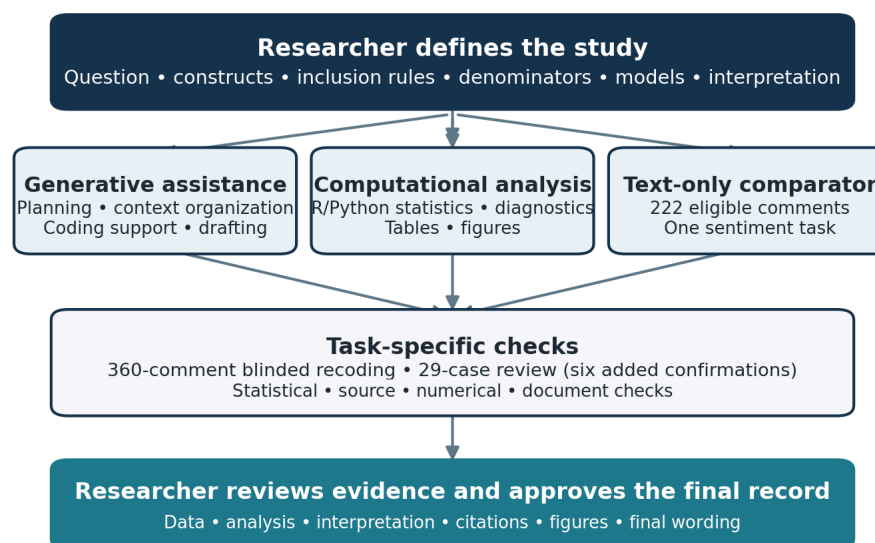
Table F1
Task-Level Assistance, Researcher Decisions, and Checks

Task area	How assistance was used	Researcher decision and task-specific check
Design and selection	Planning options, metadata-level candidate-prioritization support, and multimodal context organization	Formulated the question and set the scope, constructs, codebooks, denominators, and models; directed staged shortlisting; opened and assessed all 314 transcript/context candidates; made every final inclusion decision; selected 217 videos, of which 216 had usable direct threads and formed the linked analytical corpus
Coding and qualitative synthesis	Applied the final researcher-defined prompt and codebook to the full comment corpus; organized qualitative case material	Defined labels, boundaries, and calibration examples; reviewed the predictor-blinded reproducibility and targeted-comparison results; decided the final analytical use of the coded fields; and reviewed all 29 qualitative cases
Code, statistics, and validation	R/Python execution, diagnostics, tables, and figures	Set specifications; checked denominators, model outputs, diagnostics, and reported values
Sources, organizational mapping, and writing	Search planning, source-location and consistency checks, synthesis, wording and structural revision, and document QA	Checked source-to-claim fit; reviewed, revised, accepted, or rejected proposed wording and structure; approved the submitted citations, implications, displays, and text

Note. Generative AI means language-model assistance; the statistical analysis in R/Python and the sentiment-classifier run were separate computational activities. Each check supports only its named task. The recoding check measures AI-system run-to-run consistency under predictor blinding, not agreement between human coders. Author's process, validation, and source records.

OpenAI ChatGPT Work/Codex was used for candidate prioritization, structured coding support, qualitative organization, source-location and consistency checks, wording and structural revision, and document quality assurance. For each task, the system generated proposals or structured outputs; the researcher reviewed them, decided what entered the project, and remained responsible for the submitted design, data choices, analyses, interpretations, citations, structure, and wording. The final manuscript-revision and document-quality-assurance phase in August 2026 used GPT-5.6 Sol (gpt-5.6-sol; OpenAI, 2026). The fixed sentiment classifier was run separately from ChatGPT and from the statistical analyses in R and Python. The targeted comparison used a locally archived, quantized Open Neural Network Exchange (ONNX) implementation of Cardiff NLP's twitter-roberta-base-sentiment-latest as an external benchmark for 222 eligible comments. The classifier received only the written comment, without the linked video or thread context. Barbieri et al. (2020) document the TweetEval benchmark; Loureiro et al. (2022) and Cardiff NLP (2022) document the updated model. These sources do not document the local conversion. Research decisions, context-aware labels, and the other coded fields remained under researcher control. Figure F1 separates researcher-defined design, bounded assistance, computational analysis, the classifier comparison, task-specific checks, and final researcher approval.

Figure F1
Researcher Oversight, Task-Specific Checks, and Final Authority

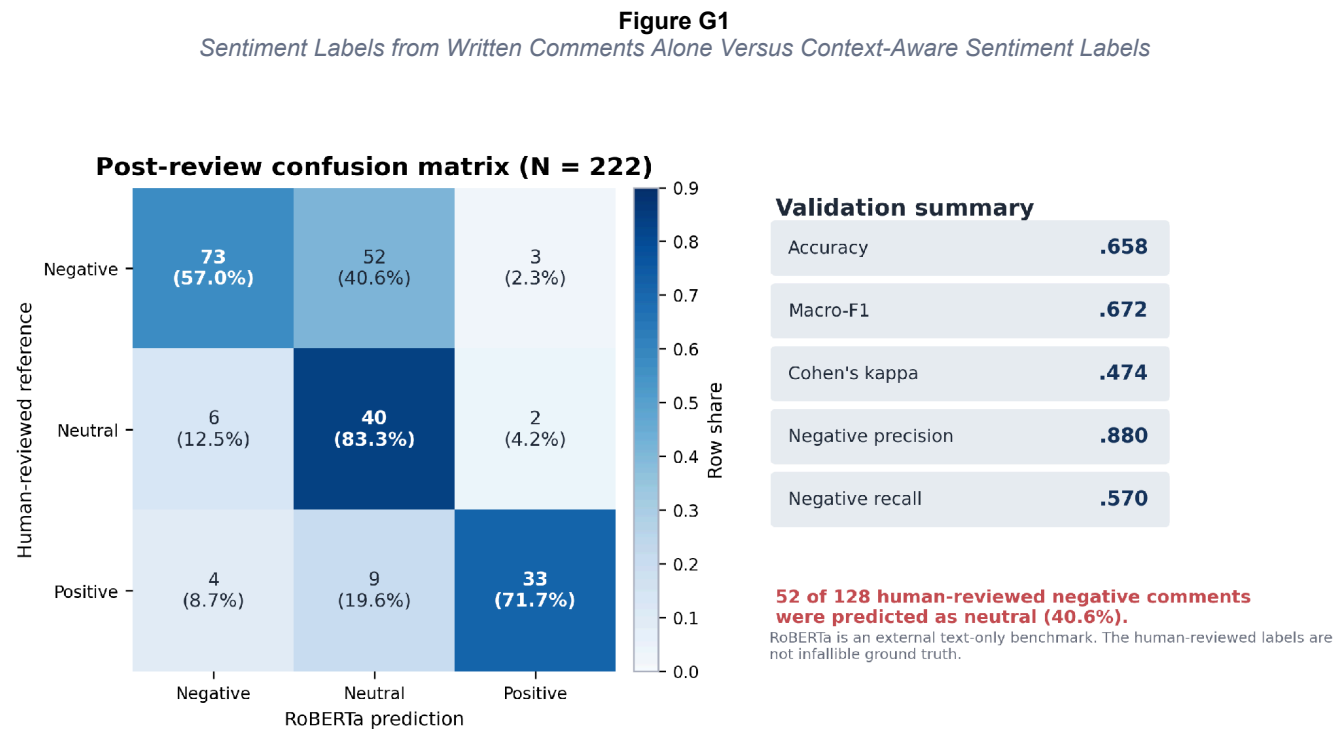


Note. Recoding covered 360 comments in two separate predictor-blinded runs by the same AI system; the targeted classifier comparison covered 222 eligible comments from a 228-comment set; and case review covered all 29 interpretations, with six additional confirmations. Other checks addressed their named outputs. The researcher approved the final record. The recoding check measures AI-system run-to-run consistency. Source: Author's process and validation records.

Data access is governed by privacy, identifier, integrity, and authorization requirements. The thesis reports aggregate outputs and privacy-screened paraphrases, except for the three short direct comments in Figure 1, whose archived wording is retained after usernames and profile images were removed. Other raw or linked TikTok material may contain exact text, captions, URLs, creator handles, stable identifiers, or linkage fields and remains restricted. The same caution applies to classifier inputs containing exact comment text. Codebooks, scripts, aggregate table or figure sources, and other nonidentifying materials may be considered only after file-level review has confirmed that sharing is authorized, any necessary identifiers have been removed or minimized, record integrity is preserved, and relevant platform and institutional requirements are met.

Appendix G. Measurement Validation and Reproducibility

This appendix reports the study's field-specific validation evidence and the reproducibility limits of the published sentiment classifier used as an external benchmark. The evidence applies only to the fields, cases, and tasks named below; it does not establish corpus-wide accuracy or validate the study's other coded fields. Figure G1 presents the targeted classifier comparison, and Table G1 summarizes the validation layers and their limits.



Note. For 222 eligible comments from the targeted 228-comment set, rows show the researcher-reviewed context-aware sentiment labels and columns show labels assigned by Cardiff NLP's fixed twitter-roberta-base-sentiment-latest classifier when it received only the written comment. Cells show counts and row percentages. Overall agreement = .6577, macro-F1 = .6715, κ = .4738, negative precision = .8795, and negative recall = .5703. The classifier provides an external benchmark for this targeted comparison; neither label set is treated as ground truth, and the comparison is not a full-corpus validation. Author's validation record.

Table G1
Validation Checks and Their Limits

Validation layer	Eligible base	Evidence	Use	Scope boundary
Predictor-blinded field recoding	Same 360 comments from 126 video groups; two independently executed, predictor-blinded runs using AI; 240 probability-sampled and 120 targeted	Agreement 77.8% ($\kappa=.738$) to 89.4% ($\kappa=.827$); primary evaluative target least stable	Diagnose run-to-run reproducibility; discrepancies flagged for review	Recodes had no automatic overwrite authority; no corpus-wide accuracy claim
Classifier/context-aware label comparison	222 eligible comments from a targeted 228-comment human-reviewed set	Overall agreement .6577; macro-F1 .6715; $\kappa=.4738$; negative precision .8795; negative recall .5703	Assess how the labels differ when the classifier receives only the written comment	Limited to this targeted comparison; neither label set is treated as ground truth
Negative-label classification pattern	128 comments assigned a negative label after context-aware review.	Using only the written comments, the classifier assigned 73 negative, 52 neutral, and three positive labels	Describe where the two label sets differed most	Classification pattern alone does not establish the correct label for each disagreement
Review of initial disagreements	76 initial three-class disagreements	Researcher context-based label retained in 75; classifier label adopted in one	Challenge and refine the targeted comparison set	Targeted adjudication, not full-corpus manual review
Classifier rerun	All 13,235 comments in the archived corpus	A local CPU rerun reproduced all 13,235 classifier labels and the reported statistics for the 222-comment comparison; recorded hashes identify the local files.	Check that the saved model and code produce the same labels and comparison statistics	Controlled inputs and logs are not submitted; the rerun does not validate label meaning

Note. Each validation layer addresses a different object. The layers are complementary and do not combine into a claim of universal accuracy, complete independent manual review, or automatic overwrite authority. Author's validation record.

The documented classifier record identifies the quantized ONNX model and tokenizer, runner, inputs, label map, preprocessing, and pinned core packages. Inference used batches of one and a maximum length of 512 tokens. URLs were normalized to “http,” eligible user mentions to “@user,” and whitespace was normalized; case, punctuation, emojis, and hashtags were retained. Two inputs reached the 512-token limit. A documented local CPU rerun reproduced all 13,235 classifier labels and the reported statistics for the 222-comment comparison; recorded file hashes identify the local files. The complete controlled inputs and execution logs are not included with the thesis, and the upstream conversion revision was not retained; the rerun is therefore a documented project reproducibility check, not a claim that the upstream conversion can be independently recreated from the thesis alone.

Table G2
AI-System Run-to-Run Agreement Across Two Separate Predictor-Blinded Coding Runs

Field	A–B agreement	κ	Pass A–final coding	Pass B–final coding
Tier	86.1%	.784	81.7%	80.8%
Direction toward the firm	89.4%	.827	76.7%	75.8%
Sentiment	87.5%	.823	70.8%	73.1%
Legitimacy	86.1%	.799	67.5%	66.9%
Credibility challenge	87.2%	.806	78.6%	77.5%
Explicit mismatch	84.7%	.756	75.8%	74.2%
Explicit expectation	88.1%	.822	76.7%	74.7%
Action-oriented wording	83.1%	.752	77.8%	77.2%
Sarcasm	89.2%	.775	83.6%	82.5%
Primary evaluative target	77.8%	.738	59.2%	62.5%

Note. The AI system recoded the same N = 360 comments in two independently executed, predictor-blinded runs. Agreement was assessed separately for each construct. The coefficients measure run-to-run consistency of the configured AI system, not independent-coder agreement. Comparisons with the final project coding were diagnostic, and discrepant recodes did not automatically overwrite existing values. Author's validation analysis.

Table G3
Execution Environments and Analytical Roles

Stage	Environment and material settings	Analytical role and boundary
Primary analysis environment	R 4.3.3; Ubuntu 20.04.6; glmmTMB 1.1.14; TMB 1.9.21; lme4 2.0-6; 14-regression main-analysis set; 217 input rows; 1,000 DHARMA simulations	Authoritative model estimates; associational interpretation.
Residual-quantile diagnostic	DHARMA 0.5.0; qgam 2.0.0; 1,000 simulations; refit = FALSE; seed = 2026; rank-transformed fitted-response predictor; quantiles .25/.50/.75	Combined p = .01655; .75-quantile p = .00552; diagnostic use in Appendix H.
Targeted verification	R 4.5.2; Windows 11; glmmTMB 1.1.14; TMB 1.9.23; lme4 2.0-6	Targeted fits converged with positive-definite Hessians; the low-memory check omitted DHARMA and did not replace the final model record.
Classifier rerun	Python 3.13.5; onnxruntime 1.22.1; tokenizers 0.21.4; local CPU; batch size 1; maximum length 512; fixed preprocessing	The rerun reproduced all 13,235 classifier labels and the reported statistics for the 222-comment comparison; independent reproduction requires the controlled files.

Note. The rows distinguish the primary-analysis environment, later residual diagnostic, targeted verification, and reported classifier rerun; the stages are not interchangeable. The 217 input rows refer to the full selected-video data frame; each fitted model used its outcome-specific eligible base. Source: retained primary-analysis, residual-diagnostic, later-verification, and classifier-rerun project records. Software citations: R Core Team (2024, 2025), Bates et al. (2015), Brooks et al. (2017), Fasiolo et al. (2021), Hartig (2026), and Kristensen et al. (2016).

Appendix Table H4 maps the 14-regression main-analysis set to its four outcome families; the diagnostic and verification stages shown here did not replace that set.

Additional Video-Field Agreement

Table G4 adds full-file occurrence profiles to the field-agreement evidence. These secondary and exploratory fields describe visible context; evidence quality, truth, harm, causal effect, and response effectiveness require separate evidence.

Table G4
Full-File Context Profiles and Boundaries

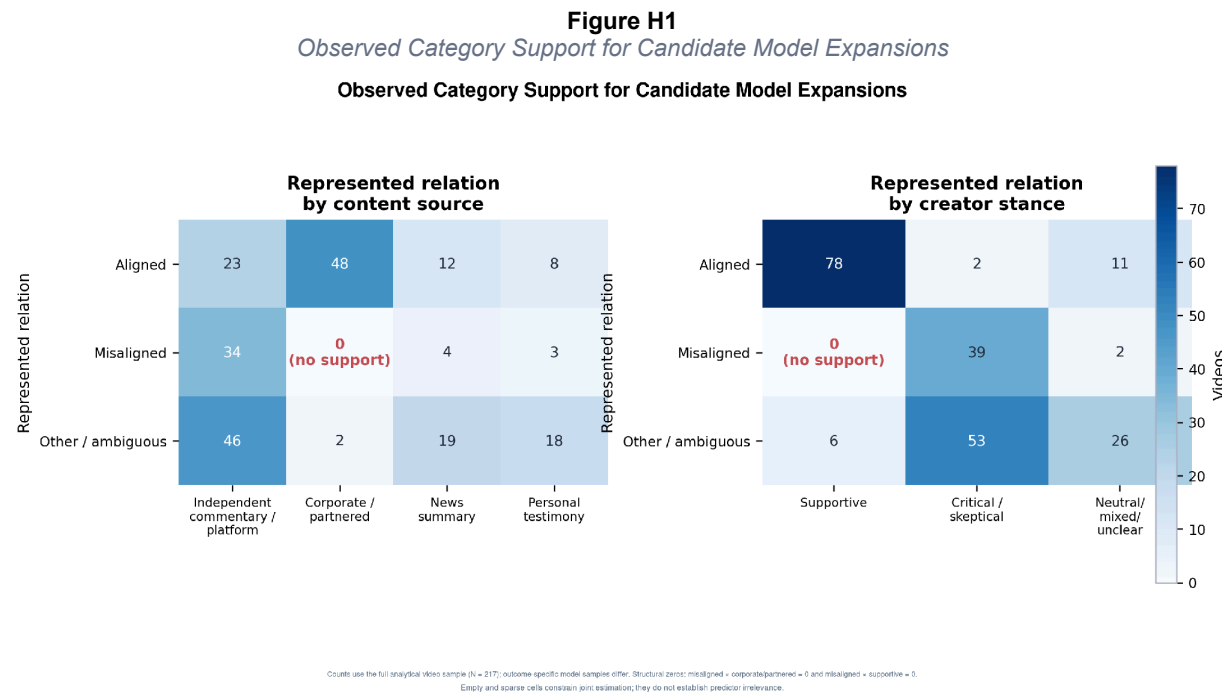
Field	Full-file profile	Agreement	Allowed use / boundary
Event specificity	Named site or incident 88; firm-specific non-site 104; industry or systemic 17; general or evergreen 7; unclear 1	Agreement 88.9%; $\kappa = .826$	Context profile and case-selection cue; no event-effect inference
Collapsed event trigger	Specific event or initiative 167; recurring or general 46; mixed or unclear 4	Agreement 86.1%; $\kappa = .718$	Context profile; no causal-salience inference
Document or report evidence	Yes 53; no 156; unclear 8	Agreement 87.5%; $\kappa = .733$	Visible evidence form; no truth, quality, or persuasion inference
Personal testimony	Yes 52; no 163; unclear 2	Agreement 91.7%; $\kappa = .818$	Visible testimony; no factual-verification inference
Expert or authority evidence	Yes 76; no 135; unclear 6	Agreement 76.4%; $\kappa = .530$	Descriptive use only
Corporate-response presence	Yes 154; no 60; unclear 3	Agreement 91.7%; $\kappa = .798$	Presence only; no response-effectiveness inference
Stakeholder burden	Present 131; absent 86	Binary agreement 86.1%	Context profile and future hypothesis; no burden-effect inference

Note. Counts use all 217 selected videos. For the listed fields, agreement compares an independent recode of 72 selected videos with the original field assignments; stakeholder burden uses its separate documented binary-coding check. Cohen's κ is reported for the six fields whose paired categorical comparison is retained; the project record does not preserve a κ for stakeholder burden. The profiles describe coded context and provide no independent effect estimate. Source: final video file and final validation record.

Appendix H. Supplementary Profiles and Model-Architecture Diagnostics

The source-adjusted claim-visibility model covered 216 videos. Explicit claim visibility versus no clear claim produced $OR = .906$, 95% CI [.630, 1.305], $p = .597$, Benjamini–Hochberg-adjusted $q = .784$; the claim-visibility block test gave $p = .096$. The residual-quantile test reported in Appendix Table G3 returned a combined $p = .01655$, with the .75-quantile component $p = .00552$. The descriptive contestability gradient remains separate; this diagnostic keeps the adjusted claim-visibility model appendix-only and prevents it from being read as a distinct supported claim-visibility association.

Figure H1 shows the source–relation and stance–relation combinations observed before Table H1 reports the corresponding candidate model expansions.



Note. Relation-by-source and relation-by-stance cross-tabulations cover all 217 videos; outcome-specific bases differ. The two structural zeros and the sparse aligned-by-critical cell constrain joint estimation. The display does not establish causal identification, causal effects, or general predictor irrelevance. Author's analysis.

Table H1 a
Integrated-Model Architecture Diagnostics

Panel A. Fit Comparison

Candidate addition	Added-block likelihood-ratio p	Comparator AIC	Candidate AIC	ΔAIC
Relation added to substantive-comment-yield architecture	.962983	1437.725	1438.445	+.720846
Relation added to conditional-firm-criticism architecture	.540466	796.658	799.428	+2.769354

Table H1 b

Panel B. Grouped Prediction and Decision Meaning

Candidate addition	Firm-grouped predictive difference per video	Decision meaning
Relation added to substantive-comment-yield architecture	+.000406	Retain the substantive-comment-yield outcome model; the larger candidate added no fit gain and predictive performance was effectively unchanged.
Relation added to conditional-firm-criticism architecture	-.023954	Retain the conditional-firm-criticism outcome model; the larger candidate added no fit gain and grouped prediction worsened.

Note. Block tests, AIC comparisons, and firm-grouped predictive differences govern the local candidate-architecture decision. They do not test focal effects, identify causality, or establish general predictor irrelevance. Author's analysis.

The support gaps and the lack of incremental fit or firm-grouped predictive gain explain why the candidate integrated predictor models in Table H1 did not replace the final outcome-specific specifications. These candidates were model-architecture checks outside the 14-regression main-analysis set; they do not establish identification or make the predictors generally irrelevant.

Table H2 then shows how conditional firm criticism can coexist with supportive creator presentation and keeps the targeted favorable-or-mixed caveat check on its separate descriptive base.

Table H2 c
Supportive-Message Contestation and Targeted Caveat Profiles

Profile item	Value	Reader-facing role
Supportive videos in full sample	84	Full-sample context
Eligible / zero firm-directed Tier-A denominator	67 / 17	Outcome eligibility
Eligible threads with any against-firm comment	55	Threads containing criticism
Majority against / tie / majority not against	37 / 2 / 28	Complete eligible partition
Eligible-video criticism rate, median / mean	57.1% / 52.7%	Video-level distribution
Pooled supportive-thread criticism	567/1,216 (46.6%)	Firm-directed Tier-A base for supportive videos
Targeted favorable-or-mixed comments	524 targeted favorable/mixed Tier-A comments; 457 classifiable; 119/457 caveated (26.0%); 338/457 not caveated; 67 unclear excluded; documented repeat-coding agreement 89%; $\kappa = .763$	Targeted coding check

Note. These are descriptive profiles. They show that supportive creator stance and against-firm comments can coexist; they do not estimate a causal creator-stance effect. The targeted favorable-or-mixed set is a separate descriptive coding check and does not estimate trust, endorsement, or corpus-wide prevalence. The repeat-coding value describes agreement in that targeted coding check and within-researcher reproducibility, not independent-coder agreement. Author's analysis.

Table H3 consolidates the full selected-video record, platform, video, and outcome characteristics so that the distinct denominators and aggregation levels can be checked together.

Table H3 a*Full Selected-Video Record, Platform, Video, and Outcome Characteristics***Panel A. Full Selected-Video Record and Platform Characteristics**

Characteristic	N	Mean or n	Median or %	IQR	Range	Notes
Firms	40	40				Final analytical sample
Videos	217	217				Full selected-video record
Videos with a usable direct-comment thread	217	216	99.5%			216 linked video-comment threads; one selected record had no usable direct thread
Unique direct comments	13,235	13,235				Replies excluded
Views	217	498,048	29,400	6,738–90,800	459.0–39,500,000	All videos
Platform comment count	217	247.7	46.0	19.0–149.0	10.0–8,424	All videos
Platform engagement rate	217	8.39%	6.24%	3.61%–12.18%	0.15%–38.43%	All videos; (likes + platform comments + shares + saves) / views
Creator followers	217	800,707	70,600	11,200–401,600	23.0–18,500,000	All videos
Observed video duration (seconds)	208	69.3	59.0	36.0–85.0	5.0–596.0	208 videos with positive observed duration; 9 zero/missing values were median-imputed to 59 seconds for modeling only
Video age (days)	217	605.4	369.0	211.0–973.0	14.0–2,212	All videos
Retrieved comments	216	61.3	26.5	12.0–89.8	3.0–200.0	216 comment-bearing videos

Table H3 b

Panel B. Video and Communication Characteristics (N = 217)

Characteristic	n	%
Source form — Independent commentary/platform material	103	47.5%
Source form — Corporate or partnered	50	23.0%
Source form — News summary	35	16.1%
Source form — Personal testimony	29	13.4%
Represented relation — Aligned	91	41.9%
Represented relation — Other/ambiguous	85	39.2%
Represented relation — Misaligned	41	18.9%
Creator stance — Critical/skeptical	94	43.3%
Creator stance — Supportive	84	38.7%
Creator stance — Neutral/mixed/unclear	39	18.0%
Diagnostic frame — Present	172	79.3%
Diagnostic frame — Absent	45	20.7%
Prognostic frame — Present	164	75.6%
Prognostic frame — Absent	53	24.4%
Motivational frame — Present	53	24.4%
Motivational frame — Absent	164	75.6%
Conduct domain — Strategic/governance/finance	65	30.0%
Conduct domain — Operational/environmental	98	45.2%
Conduct domain — Communication/claim	20	9.2%
Conduct domain — Product/service	33	15.2%
Conduct domain — Other/unclear	1	0.5%
Sustainability-claim visibility — Explicit claim	117	53.9%
Sustainability-claim visibility — Attributed/reputational	27	12.4%
Sustainability-claim visibility — No clear claim	73	33.6%
Sustainability centrality — Core	169	77.9%
Sustainability centrality — Contextual	48	22.1%

Table H3 c

Panel C. Video-Level Outcome-Rate Distributions

Characteristic	N	Mean or n	Median or %	IQR	Range	Notes
Substantive-comment yield	216	55.8%	60.4%	33.3%–80.0%	0.0%–100.0%	216 comment-bearing videos; video-level Tier-A/raw-comment rate
Conditional firm-criticism rate	186	62.0%	66.7%	38.1%–96.8%	0.0%–100.0%	186 videos with at least one substantive firm-directed comment
Context-aware negative-sentiment rate	209	43.6%	43.8%	21.4%–66.7%	0.0%–100.0%	209 videos with a valid three-class context-linked sentiment denominator
Non-firm primary-target rate	201	44.9%	50.0%	11.9%–71.4%	0.0%–100.0%	201 videos with at least one stable substantive target

Note. Continuous summaries are unweighted across eligible videos. Platform engagement rate is defined in §3.3. The duration row uses 208 positive observations; nine zero or missing values were set to the 59-second median for model controls only. Panel B uses N = 217 selected-video records. Diagnostic, prognostic, and motivational frames are overlapping binary indicators; represented conduct domains are mutually exclusive. Pooled counts are 7,302/13,235 for substantive-comment yield, 1,893/3,260 for conditional firm criticism, 4,105/8,898 for context-aware negative sentiment, and 3,146/6,406 for non-firm primary-target rate. Outcome-specific video threads are the grouped models' units; comments supply their numerators and denominators. Author's calculations.

Table H4 separates the symmetric Spearman comparison, the 14 regressions in the main-analysis set, and the separately reported F02 sensitivity. It aligns each outcome with its tested video characteristics, specification, and eligible base.

Table H4
Outcome Families, Analysis Status, and Eligible Bases

Analysis / status	Outcome or measures	Tested video characteristics	Specifications / method	Eligible base
Spearman association	Platform engagement rate ↔ substantive-comment yield	No formal dependent or independent variable	One Spearman rank correlation	216 videos; 40 firms
Main analysis · M1	Substantive-comment yield (Tier A / retrieved direct comments)	Represented relation; source form; framing; claim visibility	Six beta-binomial mixed-effects regressions: M1_R; M1_CS; M1_F; M1_F_source; M1_CV; M1_CV_source	7,302/13,235; 216 videos; 40 firms
Separately reported sensitivity · F02	Substantive-comment yield	Represented relation + source form	M1_R_source (historical F02); coefficient p reported outside the 126-test adjustment	7,302/13,235; 216 videos; 40 firms
Main analysis · M2	Conditional firm criticism	Represented relation; creator stance; represented conduct domain; source form	Four beta-binomial mixed-effects regressions: M2_R; M2_S; M2_CD; M2_CS	1,893/3,260; 186 videos; 38 firms (M2_CD: 1,893/3,256; 185 videos)
Main analysis · M3	Context-aware negative sentiment; negative-or-mixed reaction sensitivity	Represented relation; source form in the adjusted model	Three beta-binomial mixed-effects regressions: M3_C; M3_C_source; M3_CM	4,105/8,898; 209 videos; 38 firms (negative-or-mixed: 4,378/9,171)
Exploratory main analysis · M4	Non-firm primary-target rate	Represented relation	One beta-binomial mixed-effects regression: M4_RA	3,146/6,406; 201 videos; 38 firms

Note. The main-text regression analysis reports 15 specifications: 14 regressions in the main-analysis set and one separately reported source-adjusted sensitivity (M1_R_source; historical F02). Figure 4 shows the 13 M1–M3 regressions and F02; M4_RA is reported in §4.5 and Table 5. The Benjamini–Hochberg adjustment covers 126 fixed-effect coefficient tests from the 14-regression main set; $q < .05$ indicates statistical significance after accounting for multiple comparisons. Whole-characteristic likelihood-ratio p values are separate. F02 is evaluated with its coefficient p value and does not enter the 126-test adjustment. The 14 regressions used a logit link, the same five controls, and a firm random intercept. F02 used the same structure and additionally held source form constant. Descriptive profiles and other sensitivity analyses remain outside this set. Source: Author's analysis of the final analytical records.

Appendix Table H5 provides the complete display of the focal M2 and M3 contrasts for conditional firm criticism and context-aware negative sentiment. It supplements the selected main-text table and forest plot with the complete focal numerical results.

Table H5

Complete Focal-Contrast Results for Conditional Firm Criticism and Context-Aware Negative Sentiment

Model / video predictor	Contrast	OR [95% CI]	p	BH q	Overall characteristic test p
Panel A. Dependent variable: conditional firm criticism					
M2_R — represented relation	Misaligned vs aligned	1.594 [0.909, 2.795]	.10359	—	.18225
M2_R — represented relation	Other/ambiguous vs aligned	1.427 [0.903, 2.257]	.1280	—	
M2_S — creator stance	Critical/skeptical vs supportive	2.073 [1.298, 3.309]	.00227	.01362	.00948
M2_S — creator stance	Neutral/mixed/unclear vs supportive	1.693 [0.992, 2.890]	.0536	.1647	
M2_CD — represented conduct domain	Communication/claim vs operational/environmental	1.859 [0.880, 3.926]	.1042	.2371	.15159
M2_CD — represented conduct domain	Product/service vs operational/environmental	0.809 [0.429, 1.527]	.5138	.6961	
M2_CD — represented conduct domain	Strategy/governance/finance vs operational/environmental	1.402 [0.889, 2.213]	.1460	.2968	
M2_CS — source form	Corporate/partnered vs independent commentary/platform material	0.780 [0.431, 1.409]	.4099	.5955	.11007
M2_CS — source form	News summary vs independent commentary/platform material	1.460 [0.841, 2.535]	.1786	.3462	
M2_CS — source form	Personal testimony vs independent commentary/platform material	0.667 [0.389, 1.145]	.1422	.2936	

Table H5 (continued)

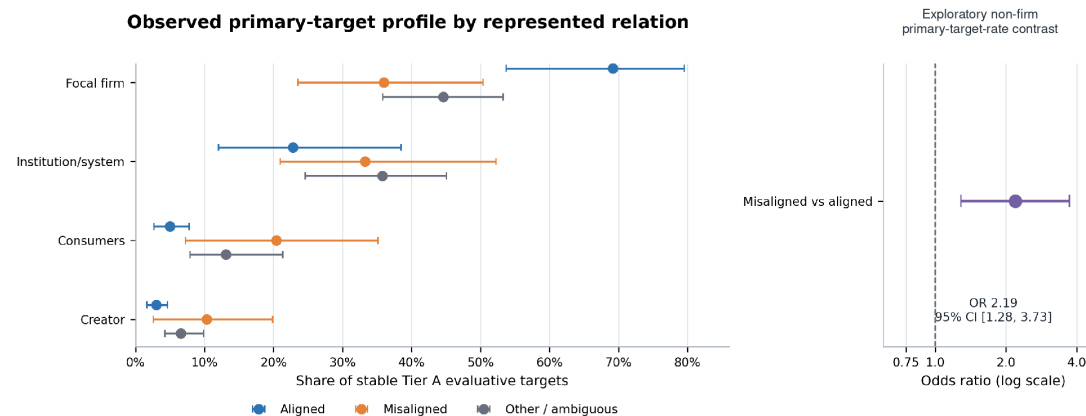
Complete Focal-Contrast Results, Panel B: Context-Aware Negative Sentiment

Model / video predictor	Contrast	OR [95% CI]	p	BH q	Overall characteristic test p
Panel B. Dependent variable: context-aware negative sentiment					
M3_C — negative sentiment	Misaligned vs aligned	1.598 [1.026, 2.491]	.03831	.12704	—
M3_C — negative sentiment	Other/ambiguous vs aligned	1.529 [1.071, 2.185]	.0195	.0703	
M3_C_source — negative sentiment, source adjusted	Misaligned vs aligned	1.782 [1.114, 2.851]	.01603	.06171	.01696
M3_C_source — negative sentiment, source adjusted	Other/ambiguous vs aligned	1.701 [1.145, 2.526]	.00848	.03880	
M3_CM — negative-or-mixed sensitivity	Misaligned vs aligned	1.596 [1.037, 2.457]	.03356	.11428	—
M3_CM — negative-or-mixed sensitivity	Other/ambiguous vs aligned	1.589 [1.125, 2.244]	.00862	.03880	

Note. OR = odds ratio; CI = confidence interval. p is the ordinary coefficient-level p value. BH q is the Benjamini–Hochberg-adjusted value from the 126 fixed-effect coefficient tests in the 14-regression analysis set. Coefficients in the primary M2_R relation model are interpreted with their ordinary p values; other coefficients are treated as statistically significant only when q < .05. A dash indicates that the archived output did not report that statistic for the row; a blank repeated-test cell means that the whole-characteristic likelihood-ratio p value is shown on the first row of the model only. Conditional-firm-criticism models use 1,893 firm-critical comments among 3,260 eligible comments across 186 videos and 38 firms, except M2_CD, which uses 1,893/3,256 across 185 videos. The negative-sentiment models use 4,105/8,898 across 209 videos and 38 firms; the negative-or-mixed sensitivity uses 4,378/9,171 across the same 209 videos and 38 firms. Whole-characteristic p values are separate likelihood-ratio tests. Author's analysis of the final archived model outputs.

Figure H2 complements the compact target summary in Figure 5 by showing both the relation-specific descriptive profile and the adjusted exploratory non-firm primary-target-rate contrast.

Figure H2
Primary Evaluative-Target Profile and Exploratory Non-Firm Primary-Target-Rate Contrast
Where is evaluative language primarily directed?



Profile denominators: 6,406 stable-target Tier A comments. The adjusted contrast concerns primary evaluative target, not causal, legal, or psychological responsibility.

Note. The descriptive panel uses the 6,406 stable Tier-A target denominator. The adjusted exploratory non-firm profile covers 201 videos from 38 firms, uses aligned relation as the reference, and shows 2,000-draw profile intervals. The descriptive and adjusted panels answer different questions; target is not causal, moral, legal, or psychological responsibility. Author's analysis.

Additional Framing Profiles

Explicit expectations appeared in 307/1,862 substantive comments under videos without prognostic framing and 1,082/5,440 with it. Action-oriented wording appeared in 1,291/7,398 context-linked Tier-A/B comments under videos without motivational framing and 554/1,780 with it. These targeted profiles describe wording patterns; they do not estimate framing effects or corpus-wide prevalence.

Appendix I. Sensitivity Analyses for the Primary Represented-Relation Comparison

Table I1 reports the complete sensitivity family for the primary represented-relation comparison. Two restricted variants have nominal $p < .05$ while the base model does not. They provide sensitivity information and do not change the primary interpretation. The crossed firm/creator random-intercept variant was singular.

The base row repeats the main conditional firm-criticism regression; the remaining rows are additional sensitivity variants outside the 14-regression main-analysis set, not additional outcome families.

Table I1
Complete Sensitivity Family for the Primary Represented-Relation Comparison

Variant	N	OR	95% CI	p	AIC / fit
Base primary model	186	1.594033	[.909210, 2.794670]	.103588	802.5716; nonsingular
Restricted to aligned and misaligned videos	110	1.892568	[1.068189, 3.353163]	.028815	486.6205; nonsingular
Denominator ≥ 5	112	1.434287	[.754593, 2.726211]	.271043	626.8520; nonsingular
Denominator ≥ 10	76	1.477323	[.726151, 3.005552]	.281529	497.9138; nonsingular
Plus source form	186	1.620921	[.890575, 2.950212]	.113948	803.2082; nonsingular
Plus retrieval query channel	186	1.684082	[.920465, 3.081197]	.090823	805.1666; nonsingular
Plus log play count	186	1.561049	[.893806, 2.726403]	.117498	801.2150; nonsingular
Plus sector family	186	1.679721	[.963185, 2.929304]	.067584	804.1384; nonsingular
Exclude missing duration	178	1.477858	[.820838, 2.660775]	.192949	762.4571; nonsingular
Core-only declared technical correction	147	1.841140	[1.043043, 3.249911]	.035264	621.6229; nonsingular
Crossed firm/creator random intercepts	186	1.594029	[.909207, 2.794664]	.103591	804.5716; singular

Note. Odds ratios compare misaligned with aligned representation. AIC values should be compared only among models fitted to the same observations; variants that change N are not directly comparable by AIC. “Singular” refers to the fitted random-effects structure. The p values are unadjusted sensitivity diagnostics and do not enter the 126-test main-analysis adjustment. The full family is shown so that the base result and its sensitivity to sample and specification choices can be assessed together. Source: Author’s analysis.

Appendix J. Retrieval Provenance, Query-Channel Diagnostics, and Replication Workflow

This appendix separates the historical July 2026 collection record from a prospective replication protocol. It reports retrieval provenance and overlap, aggregation of platform engagement rate, query-channel composition, the complete post hoc within-firm diagnostic family, and the conditions for a future collection. Query channels are treated as features of retrieval and selection, not as causal exposures. Tables J1, J3, and J4 report the corresponding retrieval and query-channel records; Figure J1 presents the prospective workflow.

J.1 Historical Retrieval, Overlap, and Deduplication

The historical screening file contained 505 rows representing 501 distinct search aliases. Three queries per historical row produced 1,515 row–query evaluations; after the four duplicated alias rows were consolidated, these corresponded to 1,503 distinct alias–query combinations. The main method reports the distinct-alias basis, while this appendix retains the historical ledger.

Table J1
Retrieval Provenance and Overlap Ledger

Ledger item	Count	Meaning
Screening-frame rows	505	Historical project screening rows, including four duplicated alias rows; 501 distinct aliases; not an authenticated official constituent count.
Historical row–query evaluations	1,515	Three queries across 505 historical rows; after duplicated alias rows were consolidated, these corresponded to 1,503 distinct alias–query combinations; not executed searches.
Signal scores 0 / 1 / 2 / 3	1,314 / 142 / 56 / 3	Historical manual pre-screen evaluations
Firms selected for metadata retrieval / planned searches	43 / 110	Selection and within-firm query-scheduling rule
Executed query strings	111	Archived executed strings; distinct from the 110 planned search routes; the retained record does not document the one-string difference.
Raw query–video rows	5,424	All query traces preserved
Unique exact URLs	5,210	URL-level consolidation basis
Excess provenance rows	214	Raw rows minus unique URLs
URLs under multiple exact queries	115	Many-to-many membership retained
Same terminal channel across aliases	105	Alias overlap without channel change
Cross-channel URLs	10	Nine in two channels; one in all three
Environment–greenwashing overlap	1	Pair count; triple-channel URL contributes where applicable
Environment–sustainability overlap	4	Pair count; triple-channel URL contributes where applicable
Greenwashing–sustainability overlap	7	Pair count; triple-channel URL contributes where applicable
Final assigned videos	113 / 75 / 29	Sustainability / environment / greenwashing; sums to 217
Final videos with multiple traces	4	Three same-channel across aliases; one cross-channel
Final-firm trace restriction	5,124 / 105	Derived rows / query strings; not a second retrieval

Note. Exact-query membership was retained before URL consolidation, and each selected video entered analysis once under one recorded assignment. Pair counts retain the one URL found through all three channels. Documented alias consolidation and query corrections are included in the ledger. Author's retrieval record.

The project record states that metadata retrieval ran on 3–4 July 2026 with clockworks/tiktok-scraper, using video-only, all-time, most-relevant retrieval with a standard cap of 50; comments, subtitles, related retrieval, and media downloads were not requested at that stage. The archive contains ten exports. Current documentation identifies the tool and present input-field semantics but cannot establish the July 2026 build, ranking, availability, returned order, or corpus (Clockworks, n.d.-b). A live rerun would be a separately specified replication attempt.

The manual pre-screen used scores from 0 to 3. A firm entered metadata retrieval when at least one record–query evaluation scored 2 or 3. For selected firms, query forms scored 1, 2, or 3 were scheduled; score-0 forms normally were not. The historical record does not establish a fixed manual scan depth for pre-retrieval scoring.

Six comment exports contained 15,647 rows: 15,646 valid comments and one scraper-error row across 240 video IDs. Selected-video restriction, stable identifiers, documented filtering, whitespace normalization, and deduplication produced 13,235 unique direct comments across 216 selected videos; one of 217 selected videos had no usable direct thread. Historical settings were commentsPerPost = 0, topLevelCommentsPerPost = 200, and maxRepliesPerComment = 0, with 3–200 realized direct comments. Replies were excluded and 200 was a collection maximum.

J.2 Project Platform Engagement Rate Distribution and Comparability

Table J2
Project Platform Engagement Rate Distribution and Comparability

Project statistic	Value	Scope
Pooled platform engagement rate	4.39%	Pooled numerator counts divided by pooled views
Unweighted video-level mean	8.39%	217-video selected-corpus distribution
Unweighted video-level median	6.24%	217-video selected-corpus distribution
Unweighted video-level IQR	3.61%–12.18%	217-video selected-corpus distribution

Note. The pooled rate divides pooled likes, platform-displayed comments, shares, and saves by pooled views. The unweighted summaries describe the 217-video distribution. These aggregation methods answer different descriptive questions. Source: Author's calculations from the final selected-video record.

Pooled counts produced a platform engagement rate of 4.39%. Across 217 selected videos, the unweighted mean was 8.39%, the median 6.24%, and the IQR 3.61%–12.18%. No external benchmark was verified as comparable in components, timing, aggregation, population, and corpus design; these values therefore characterize only the selected corpus.

J.3 Query-Channel Composition and Outcome Ecology

Table J3

Query-Channel Composition and Selected Descriptive Outcomes

Composition or outcome	Sustainability	Environment	Greenwashing
Final assigned videos	113	75	29
Critical/skeptical	31.0%	48.0%	79.3%
Represented relation — Misaligned	9.7%	16.0%	62.1%
Source form— <i>independent commentary/platform material</i>	42.5%	46.7%	69.0%
Source form— <i>corporate/partnered</i>	34.5%	12.0%	6.9%
Represented conduct domain — <i>Communication/claim</i>	6.2%	5.3%	58.6%
Explicit claim	58.4%	40.0%	72.4%
Diagnostic frame	72.6%	81.3%	100.0%
Mean platform engagement rate	7.96%	8.32%	10.25%
Median platform engagement rate	5.68%	6.35%	10.80%
Pooled substantive-comment yield	48.2%	62.8%	61.1%
Pooled conditional firm criticism	61.5%	51.8%	69.1%
Pooled context-aware negative sentiment	42.7%	46.6%	58.3%
Pooled non-firm primary-target rate	45.2%	49.1%	65.1%

Note. Video-characteristic percentages use the channel-specific assigned-video bases (113 sustainability, 75 environment, and 29 greenwashing). Pooled outcomes use their outcome-specific eligible comment bases. Channel differences describe retrieval-and-selection ecology and do not estimate keyword effects. Author's analysis.

Creator stance, represented relation, source form, represented conduct domain, claim visibility, and diagnostic framing vary substantially across retrieval channels. These profiles document retrieval-and-selection ecology and make query wording unsuitable as a causal exposure.

J.4 Complete Post Hoc Within-Firm Query-Channel Diagnostic Family

Table J4

Complete Post Hoc Within-Firm Query-Channel Diagnostic Family

Panel A. Paired Descriptive Estimates

Outcome	Pair A–B	N firms	Mean A	Mean B	Median Δ
Platform engagement rate	G–S	13	0.100	0.070	0.033
Platform engagement rate	G–E	10	0.092	0.089	0.011
Platform engagement rate	E–S	22	0.089	0.085	-0.006
Substantive-comment yield	G–S	13	0.632	0.577	0.039
Substantive-comment yield	G–E	10	0.658	0.561	0.060
Substantive-comment yield	E–S	22	0.552	0.546	0.027
Conditional firm criticism	G–S	12	0.730	0.635	0.096
Conditional firm criticism	G–E	9	0.708	0.647	0.179
Conditional firm criticism	E–S	20	0.597	0.577	0.058
Context-aware negative sentiment	G–S	13	0.508	0.407	0.100
Context-aware negative sentiment	G–E	10	0.490	0.437	0.047
Context-aware negative sentiment	E–S	21	0.470	0.364	0.105
Non-firm primary-target rate	G–S	13	0.580	0.295	0.195
Non-firm primary-target rate	G–E	10	0.584	0.511	0.117
Non-firm primary-target rate	E–S	21	0.507	0.379	0.096

Table J4 (continued)

Panel B. Test Statistics and Multiplicity Decision

Outcome	Pair A–B	W	p	BH q	Statistically significant after BH adjustment
Platform engagement rate	G–S	18	0.05737	0.3330	No
Platform engagement rate	G–E	19	0.43164	0.6152	No
Platform engagement rate	E–S	108	0.56607	0.6532	No
Substantive-comment yield	G–S	34	0.45483	0.6152	No
Substantive-comment yield	G–E	15	0.23242	0.6152	No
Substantive-comment yield	E–S	122	0.89874	0.8987	No
Conditional firm criticism	G–S	26	0.33936	0.6152	No
Conditional firm criticism	G–E	15	0.42578	0.6152	No
Conditional firm criticism	E–S	95	0.72851	0.7805	No
Context-aware negative sentiment	G–S	31	0.33960	0.6152	No
Context-aware negative sentiment	G–E	20	0.49219	0.6152	No
Context-aware negative sentiment	E–S	66	0.08880	0.3330	No
Non-firm primary-target rate	G–S	5	0.00244	0.0366	Yes
Non-firm primary-target rate	G–E	20	0.49219	0.6152	No
Non-firm primary-target rate	E–S	66	0.08880	0.3330	No

Note. G = greenwashing, E = environment, and S = sustainability. Means are firm-level means among firms represented in both compared channels; Δ is A minus B. W is the paired Wilcoxon signed-rank statistic; p values are unadjusted, and BH q values adjust the complete 15-comparison family. Source: Author's post hoc analysis of the final query-channel diagnostic outputs.

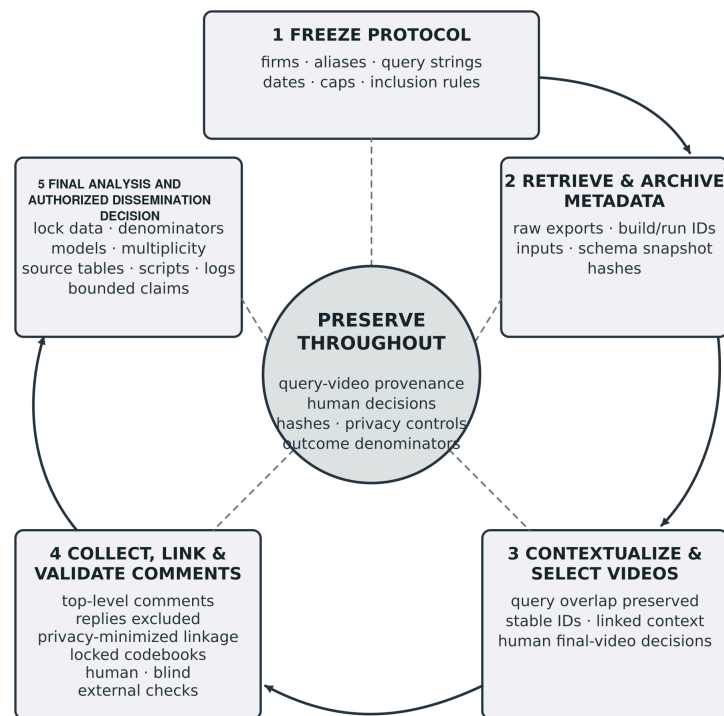
One paired descriptive difference met the Benjamini–Hochberg-adjusted criterion: non-firm primary-target rate for greenwashing versus sustainability, with 13 paired firms, means .580 and .295, median paired difference .195, $W = 5$, $p = .00244$, and BH $q = .0366$. Firm pairing reduces between-firm composition, while source, stance, issue, relation, and retrieval selection remain plausible explanations. The result describes query-channel ecology rather than a keyword effect.

J.5 Prospective Efficient Replication Workflow

A prospective collection can configure the current schema for 200 total comments, 200 top-level comments, and zero replies, subject to an immediate schema check confirming omission and pagination behavior (Clockworks, n.d.-a). This choice is distinct from the historical 0/200/0 record. A new live collection can reapply the five structured phases as a distinct prospective replication; the July 2026 Actor implementation, TikTok ranking, availability state, and returned corpus remain historical states.

Any prediction extension should define the future outcome, use a temporal holdout, prevent leakage, assess calibration and drift, and set error costs in advance.

Figure J1
Five-Phase Prospective Structured Replication Workflow



Note. The figure is an author-designed prospective workflow, not an evaluated intervention. Provenance, decisions, hashes, privacy controls, and denominator gates persist across protocol definition, metadata retrieval, video selection, comment construction and validation, and final analysis and an authorized dissemination decision. Any dissemination remains conditional on privacy review and authorization. Author's process design.

Appendix K. Possible Routes for Sustainability Strategy Review

Table K1 extends the proposed strategy use in §5.3 by stating the additional evidence required, the limits of interpretation, and the evidentiary status of each signal in this study. The routes remain illustrative and require prospective testing.

Table K1
Evidence Requirements and Interpretation Limits

Observed signal	Evidence needed	Limit of interpretation	Evidence in this study
Platform engagement rate versus substantive-comment yield	Capture completeness, exact issue, materiality, and stakeholder evidence outside the corpus	Visibility alone does not establish importance, approval, escalation, early warning, or business impact	Descriptive pattern in selected threads; proposed route for future evaluation
Claim, conduct, or credibility challenge	Approved wording, substantiation, assurance, operational records, and affected-actor evidence	Observed challenge does not itself establish greenwashing, deception, misconduct, legality, or liability	Coded discourse in selected threads; external assessment required
Source form, creator stance, or framing	Source or partnership disclosure, issue mix, audience or exposure context, and primary evidence	The associations do not establish persuasion, campaign performance, creator effects, or audience belief	Supported or qualified association only as stated
Context-aware negative sentiment versus conditional firm criticism	Video and thread context, stable target, issue facts, and external impact evidence	Negative language does not by itself establish firm blame, fault, trust loss, or representative sentiment	Distinct eligible bases; negative-sentiment association qualified
Primary evaluative target, explicit expectation, and action-oriented wording	Authority, contribution, feasibility, jurisdiction, affected-party evidence, and the wording of any explicit expectation or action	Target coding identifies the actor being evaluated. External evidence, relevant standards, and authorized judgment are needed to assess responsibility, control, and who can act.	Descriptive target and expectation/action-wording input for a subsequent evidence question; exploratory relation comparison
Data and coding quality	Eligible denominator, codebook, disagreement review, category support, and reproducibility record	Model agreement alone does not establish identification, ground truth, or automated truth	Methodological safeguard supporting review

Note. Author synthesis based on the final findings. The evidence routes are illustrative rather than validated decision rules; their practical value requires prospective testing.